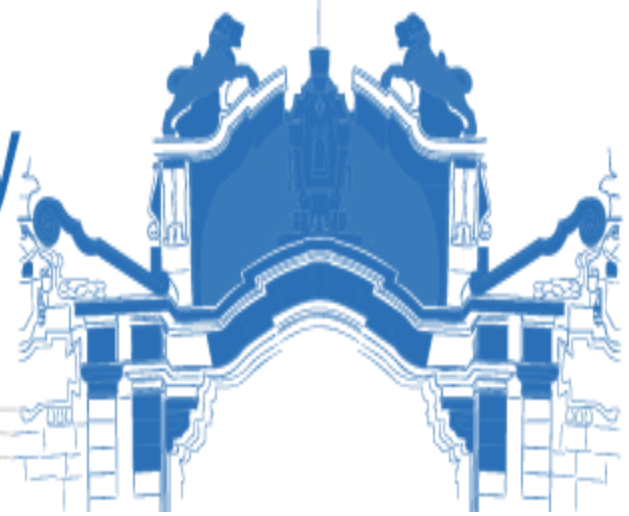




Addis Ababa University
አዲስ አበባ ዩኒቨርሲቲ

SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Physiology of Respiratory system

Fikremariam. A (Medical physiology)

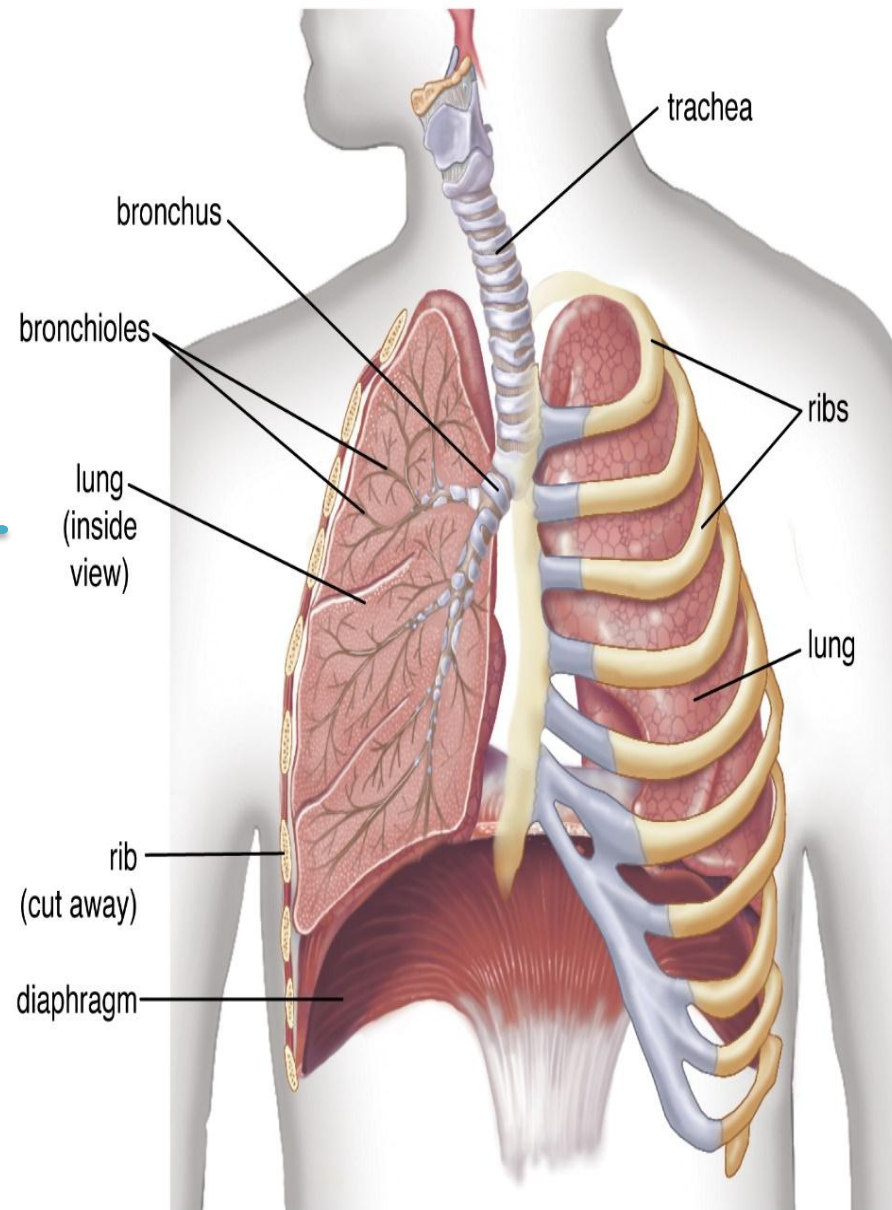
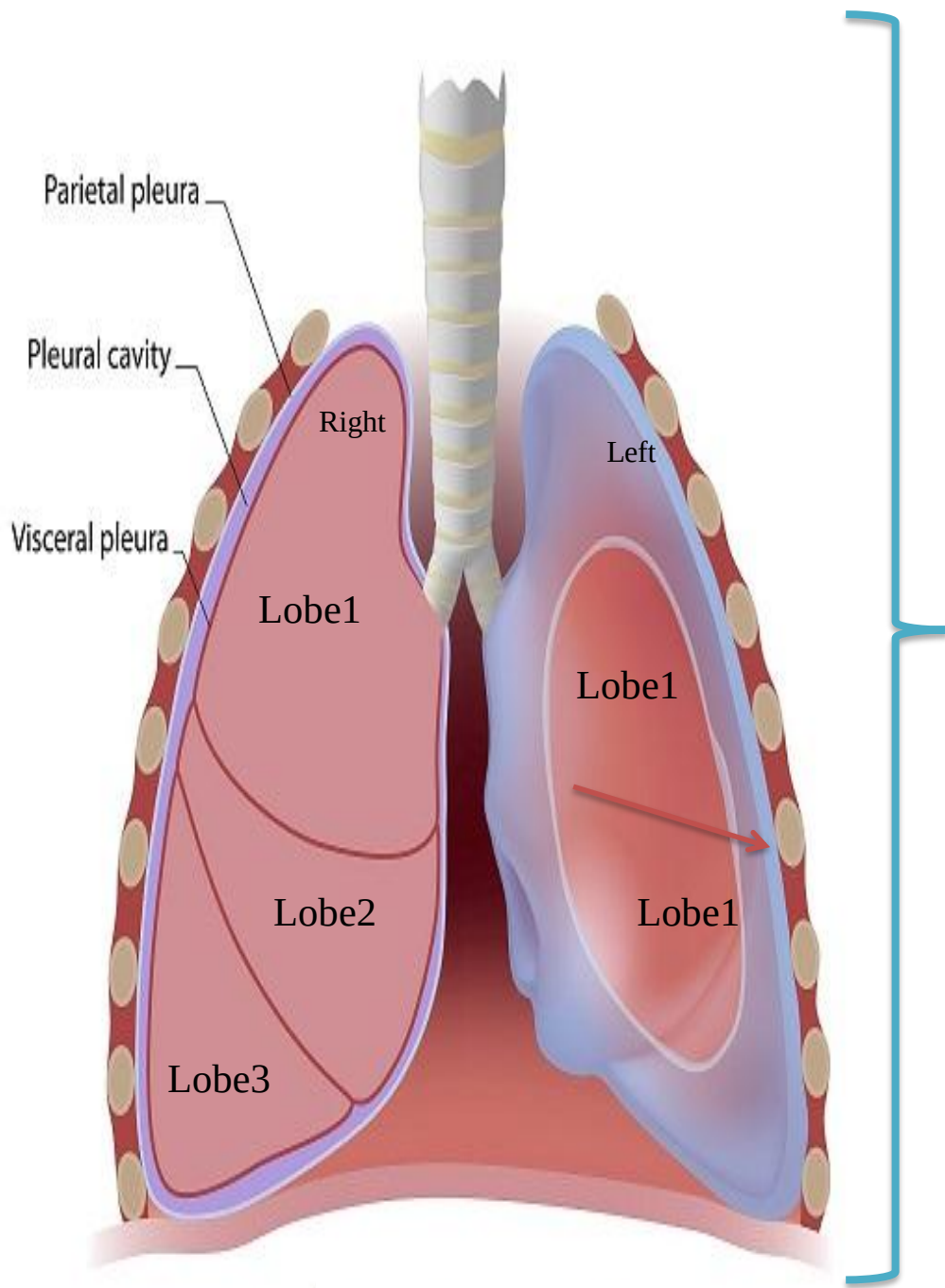
Oct,26,2016 E.C

Objectives

At the end of this chapter, students should be able to: explain about

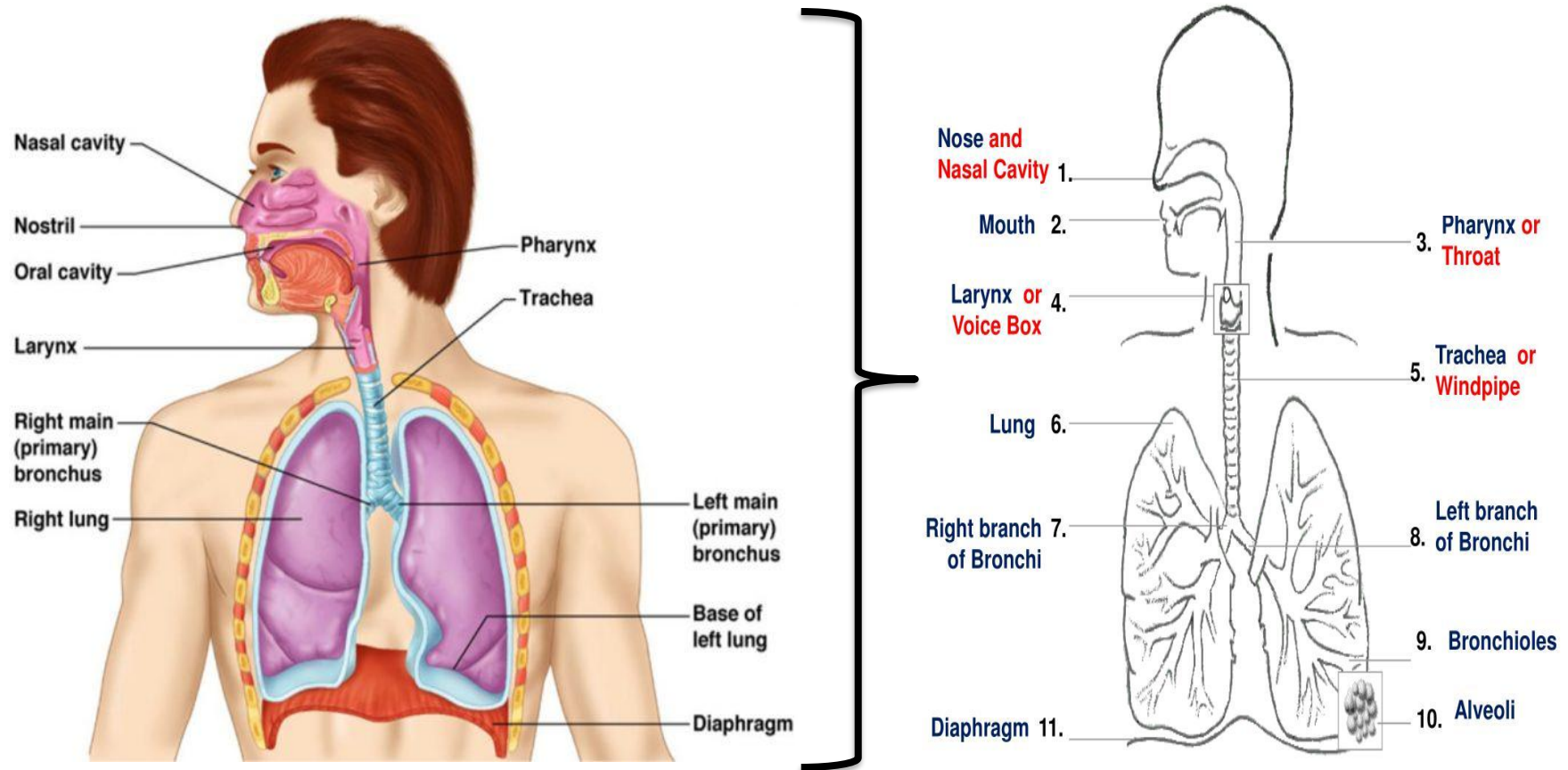
- ❑ Respiratory System (RS)
- ❑ Principal Organs of RS
 - ✓ Anatomical (Structural) &
 - ✓ Physiological (Functional) respiratory tract division
 - ✓ Functions of major organs of RS
- ❑ Functions of RS [2]
 - ✓ 1.Non-R-function
 - ✓ 2.R-function
- ❑ Respiration [Def/n]
 - ✓ Types of Respiration (internal & exter)
- ❑ Mechanics of Respiration (inspire & expire)
 - ✓ Boyle's law [$P \propto 1/V$, meaning $\uparrow P \Rightarrow \downarrow V$]
- ❑ Respiratory Muscle
 - ✓ Muscles of Inspiration
 - ✓ Muscles of Expiration

} Phases of Respiration
- ❑ Lung Volume & Capacity
 - Pulmonary Ventilation Vs Alveolar Ventilation
- ❑ Gases Exchange Or Diffusion
 - ✓ At pulmonary and Systemic Capillary
 - ✓ Factors affecting Diffusion [Exchange]
- ❑ Gases Transportation [O_2 & CO_2]
- ❑ Regulation of Respiration [2]
- ❑ Clinical correlate of Respiration



Anatomy of Respiratory System [The Principal Organs]

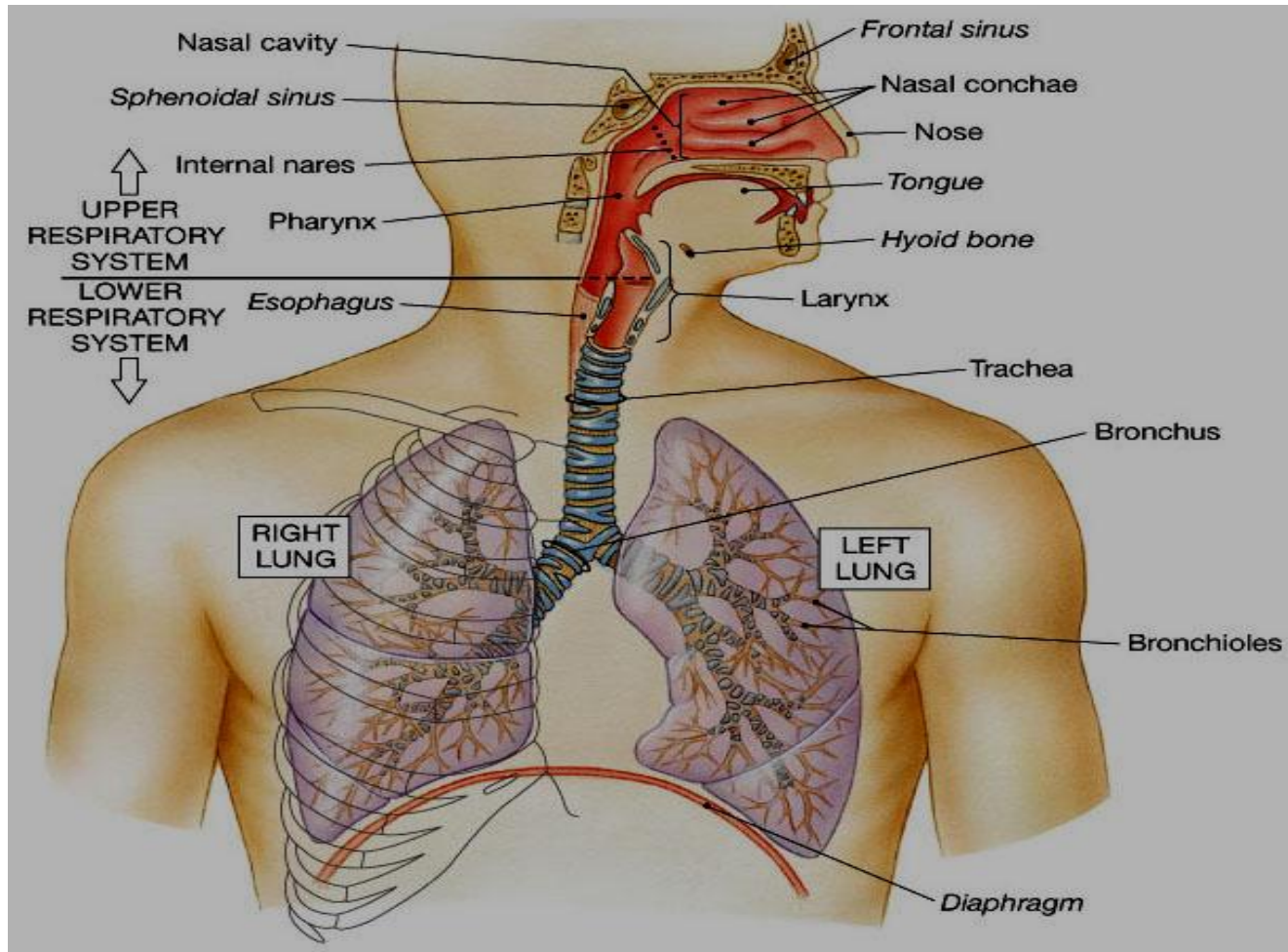
- ❑ Respiratory system air pathway consists of
- ❑ Nasal cavities or oral cavity → Pharynx → Larynx → Trachea → 1° bronchi (Rt & Lt) → 2° bronchi → 3° bronchi → Terminal bronchioles → **Respiratory bronchioles** → **Alveolar ducts** → **Alveolar sac** → **Alveoli [300M]**.



Classification of Respiratory System Structures



1. Anatomically &
2. Physiologically [Functionally]



Components of RS

Cont'd

❖ **Anatomically** ➡ the air passages of the 'RS' divided into 2 parts:

1. Upper respiratory tract

- ✓ Nose
- ↓
- ✓ Nasal cavity
- ↓
- ✓ Paranasal sinuses
- ↓
- ✓ Pharynx (throat)
- ↓
- ✓ Larynx (voice box)

2. Lower respiratory tract

- ✓ Trachea (wind pipe)
- ↓
- ✓ Bronchi [1°2°3°]
- ↓
- ✓ Terminal bronchioles
- ↓
- ✓ **Respiratory bronchioles**
- ↓
- ✓ **Alveolar ducts and**
- ↓
- ✓ **Alveoli of the lungs**

Upper respiratory tract

Nasal cavity

Pharynx

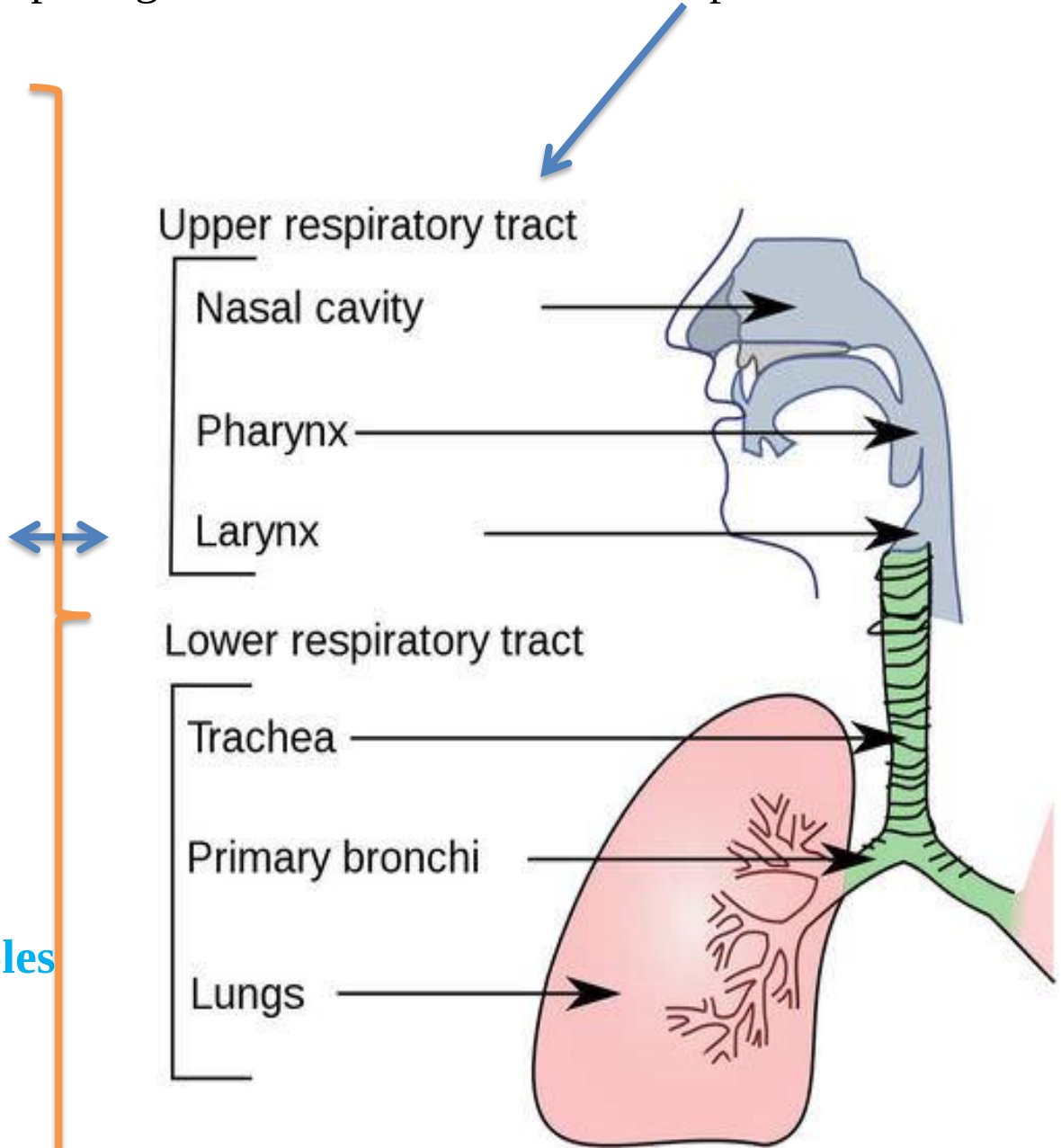
Larynx

Lower respiratory tract

Trachea

Primary bronchi

Lungs



Cont'd

❖ **Physiologically** ➡ [functionally] divided into 2 zones:

1. The Conducting zones

- ✓ Simply for passage of air, not for gas exchange & their wall is thick
- ✓ From the **nasal cavity** up to →→The **terminal** bronchioles

Function of conducting zones [in addition to as passage ways]

- ✓ To filter, clean, warm, and humidify the inspired air. &
- ✓ Secrete mucus to trap particles in the inspired air.

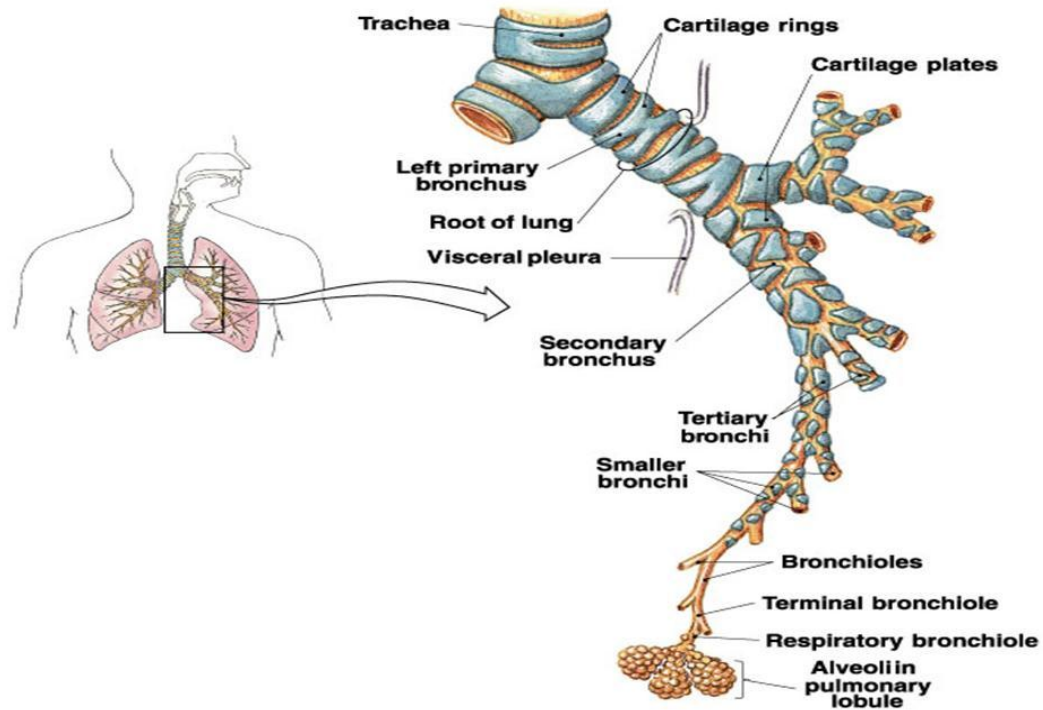
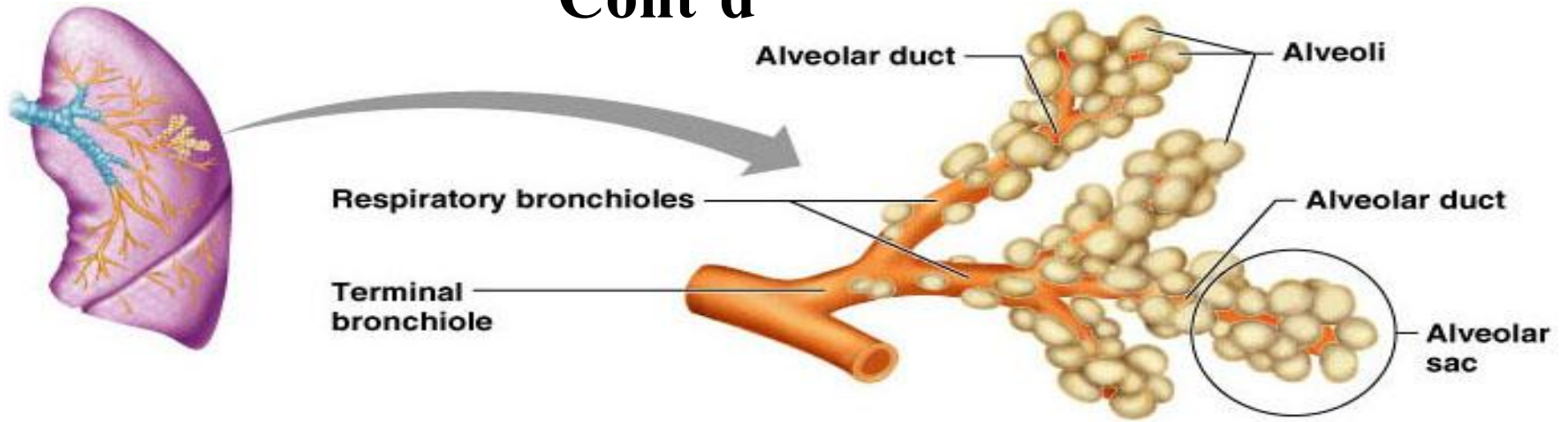
2.The Respiratory zones [Exchange zone]

- ✓ Involved in gas exchange b/c their wall is thin & Accounts for most of the lungs' volume

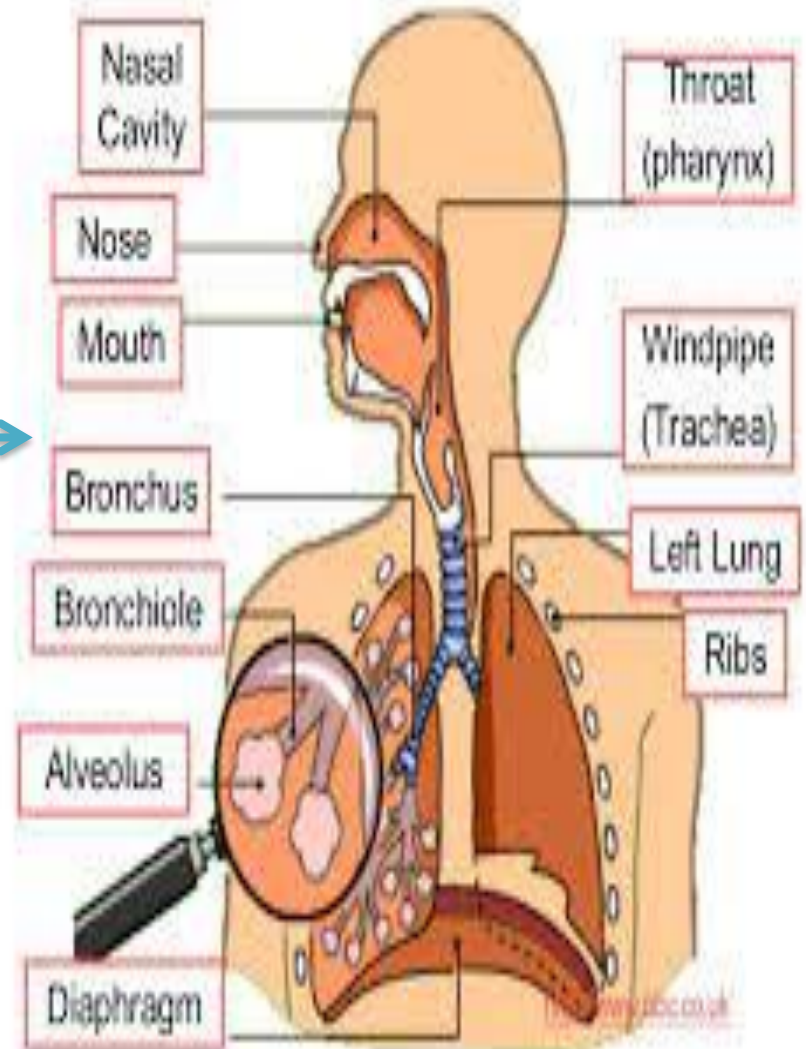
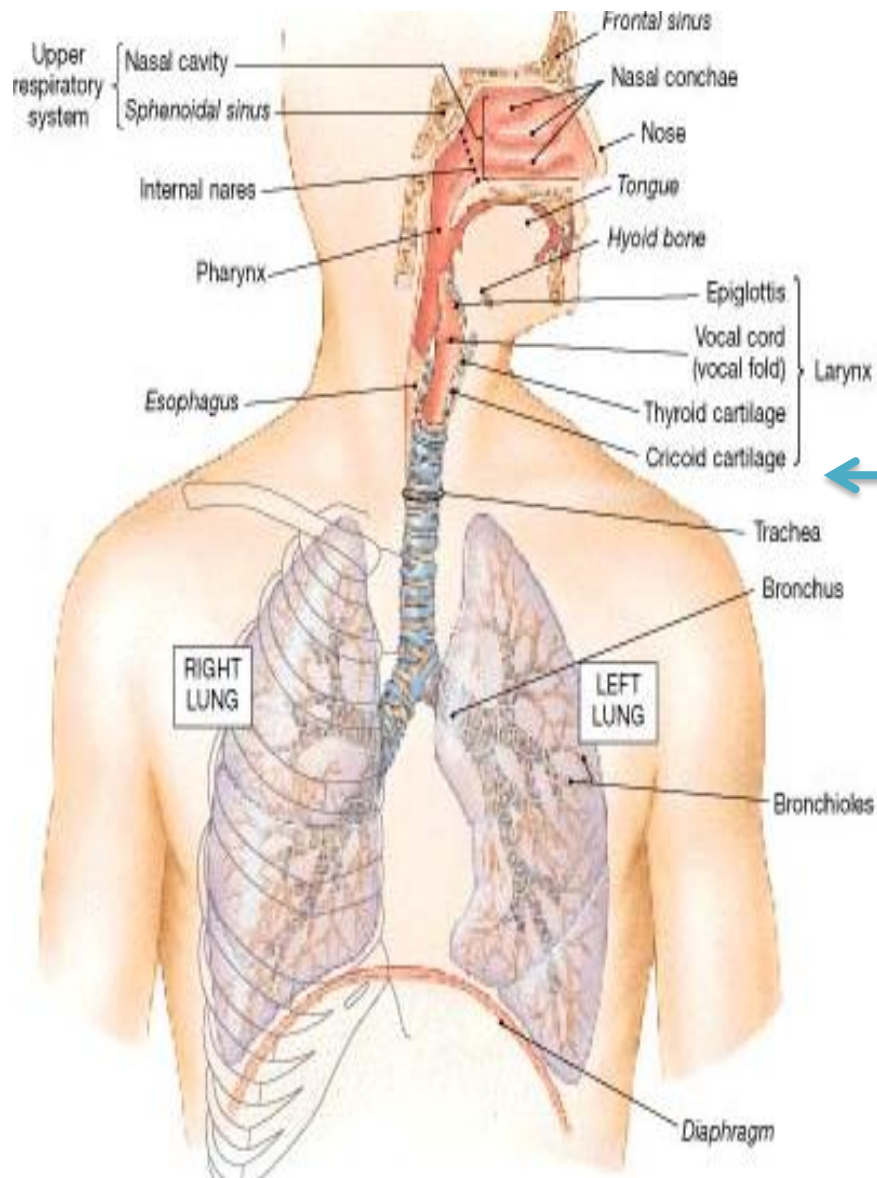
✓ Consists of

- ✓ Respiratory bronchioles →→Alveolar ducts→ alveolar sacs → **about 300 million** Alveoli
- ✓ They are the main sites of gas exchange b/n Alveolar air and blood capillary
- ✓ Provide **tremendous surface area for gas exchange**

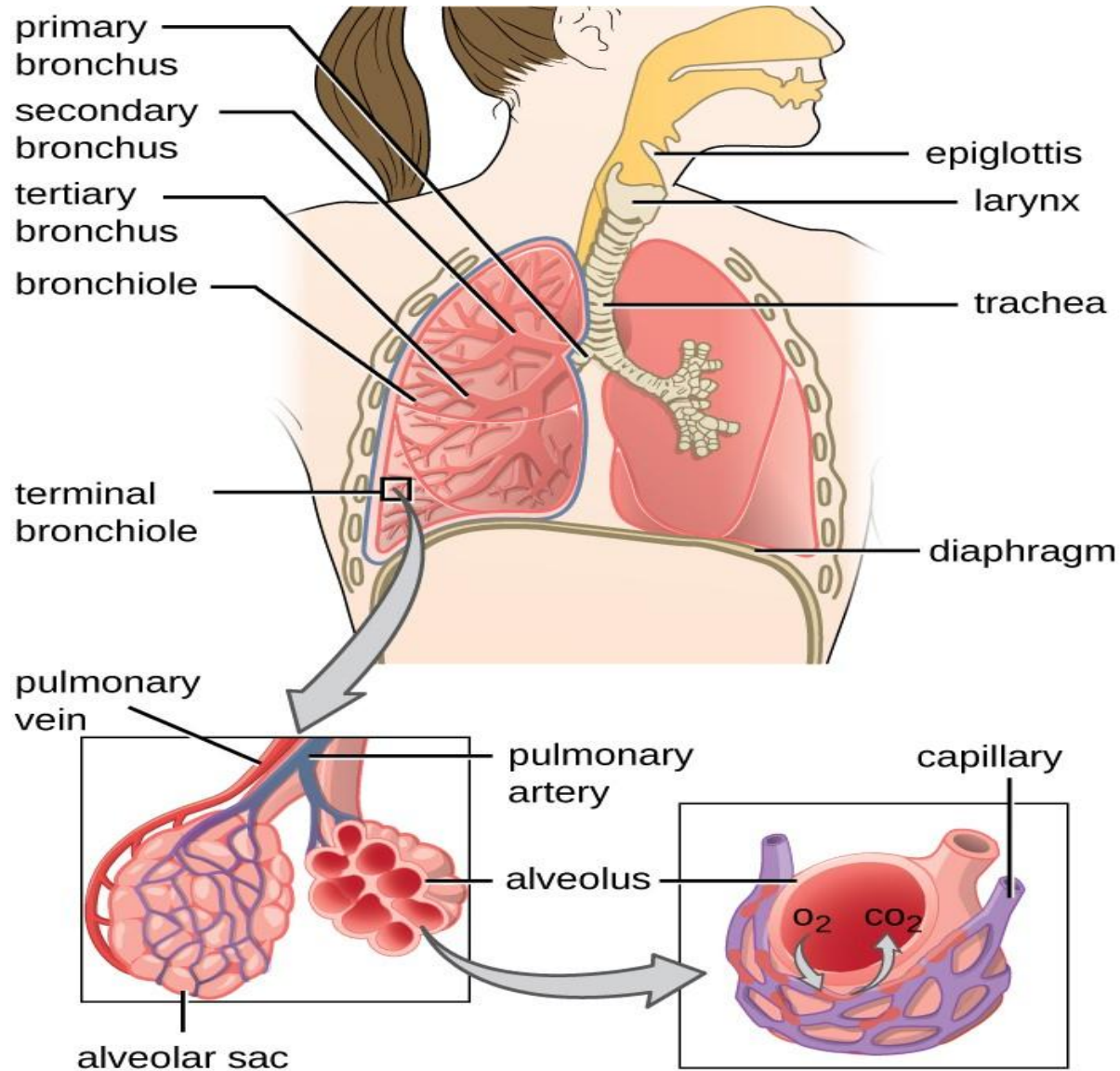
Cont'd



Functions of Major Organs of RS



Functions of Major...



Functions of Major...

1. Functions of nose and the nasal cavity [canal] entirety:

- ❑ The primary passageway for air entering the respiratory system
 - ✓ Lined with cilia, hair like tiny projections & nasal mucosa
 - ✓ Has a rich [high] blood supply and secretions
 - ✓ Warming ,filtering, clean and humidifying or moistening [conditioning of air]
 - ✓ Resonating chamber for speech
 - ✓ Houses olfactory receptors (sense of smell)

Functions of Major...

Nose



Features:

- ✓ Has External nares (nostrils) conduct air into vestibule
- ✓ Vestibule is → space lined with hairs (filter particles), leads to nasal cavity

Nasal Cavity

- ✓ From nostrils to the internal nares
- ✓ Divided into right and left by nasal septum
- ✓ Superior portion has olfactory epithelium
- ✓ Has Nasal conchae (superior, middle, inferior)

Functions of Major...

Nasal conchae:

❑ Cause air to **swirl [move around]**

- ✓ Increase likelihood of trapping foreign material in mucus
- ✓ Provide time for smell detection

Nasal Mucosa:

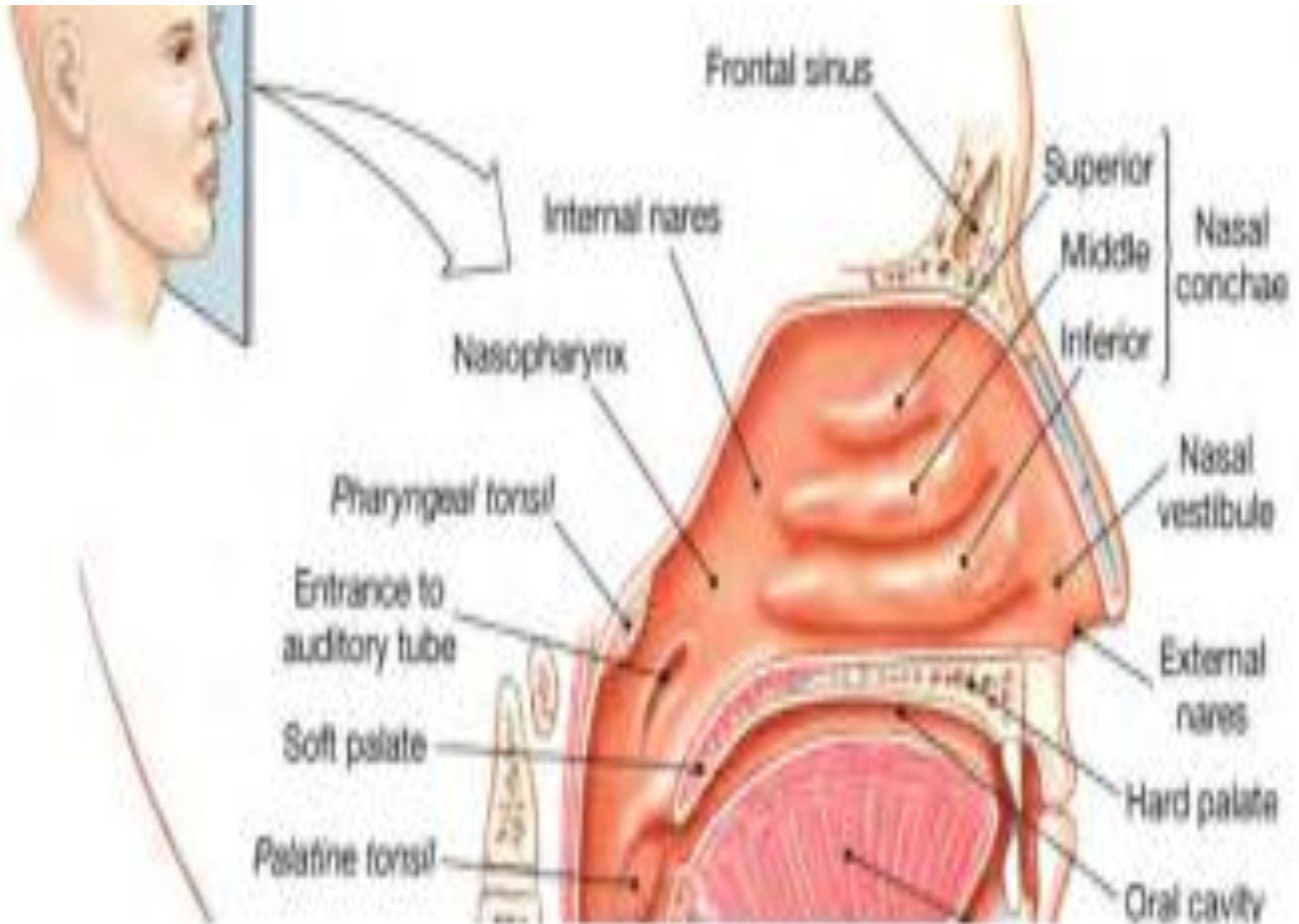
- ✓ Contains **goblet cells** ➡ secrete mucus ➡ moistens the air and traps dust particles
- ✓ Has large superficial blood supply ➡ for **warm & moisten air**.

Olfactory organs: are receptor cells for the sense of smell.

Paranasal sinuses ➡ they are lined with mucous membrane

➡ Aid in warming/moistening air

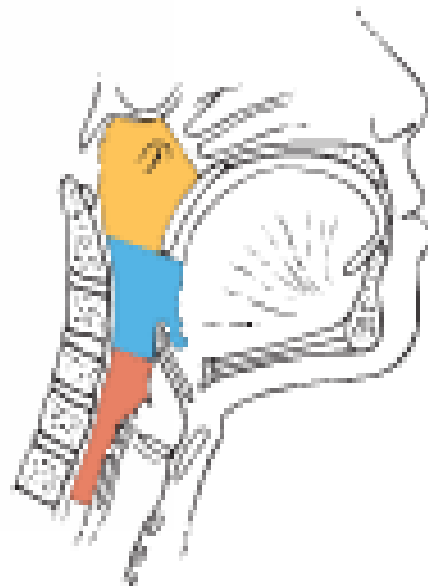
Functions of Major...



Functions of Major...

2.Pharynx (Throat) ➡ tube that delivers air from mouth & nose to the **trachea**.

- ✓ Serves as both respiratory and digestive functions.
- ✓ It breaks into **Esophagus** (used for food and fluid passage) & **Larynx**.



-  Nasopharynx
-  Oropharynx
-  Laryngopharynx

Regions of the pharynx



3 parts of Pharynx

Functions of Major...

3.Larynx [voice box it contains glottis]

- ❑ A hollow, tubular structure connected to the top of the windpipe (trachea)
- ✓ It contains the vocal cords [folds] and functions as a **voice box** for Voice production [phonation].
- ✓ During sound production, the vocal cords close together and vibrate as air expelled from the lungs passes b/n them.
- ❑ **It protects the passage of food** into the lower respiratory tracts.
 - ✓ Act as adjustment to route food and air properly [by Epiglottis]
 - ✓ **Epiglottis**- elastic cartilage flap that covers **glottis** (opening from laryngopharynx to trachea) during swallowing.

Functions of Major...

4.The Trachea

- ❑ It is located anterior to the esophagus
- ❑ Surrounded by C -shaped cartilages
- ❑ Such cartilages extend five sixth [5/6th] of the way around trachea
 - ✓ Protect the tracheal wall and prevent it from collapse
 - ✓ Keep the trachea open to allow easy passage of air
- ❑ BUT no cartilage at the back side of trachea
- ❑ What is its advantage?
 - ✓ Since esophagus is behind the trachea, absence of cartilage at the back side of trachea allows easy peristaltic movement of esophagus during deglutition of a food.
 - ✓ Trachea contains hair like called cilia :filter air before reaches to the lungs

5.Trachea bifurcates in to 2 primary **bronchi: Rt & Lt bronchi**

- ✓ Bronchi has less extensive curved cartilages
- ✓ Primary bronchus branches to secondary then tertiary bronchi ➡➡ **bronchioles**

Functions of Major...

6. Bronchioles



- ❑ Their wall **lack cartilaginous** supports
- ❑ BUT have a well developed layer of smooth muscle in their walls enables them to dilate or constrict.
- ❑ The ANS regulates the activity of this smooth muscle layer
 - ✓ Alter the diameter of the bronchioles
- ❑ Sympathetic activation ➡ bronchodilation &
- ❑ P.S stimulation ➡ a reduction of airways diameter ➡ bronchoconstriction

❖ Each bronchiole divides into ➡ terminal bronchiole ➡ Each terminal bronchiole ➡ to more smaller **respiratory bronchioles** ➡ Alveoli [300×10^6]

Functions of Major...

The Exchange zone

- ❖ Includes respiratory bronchioles and lungs (Alveoli)

7.Alveoli:

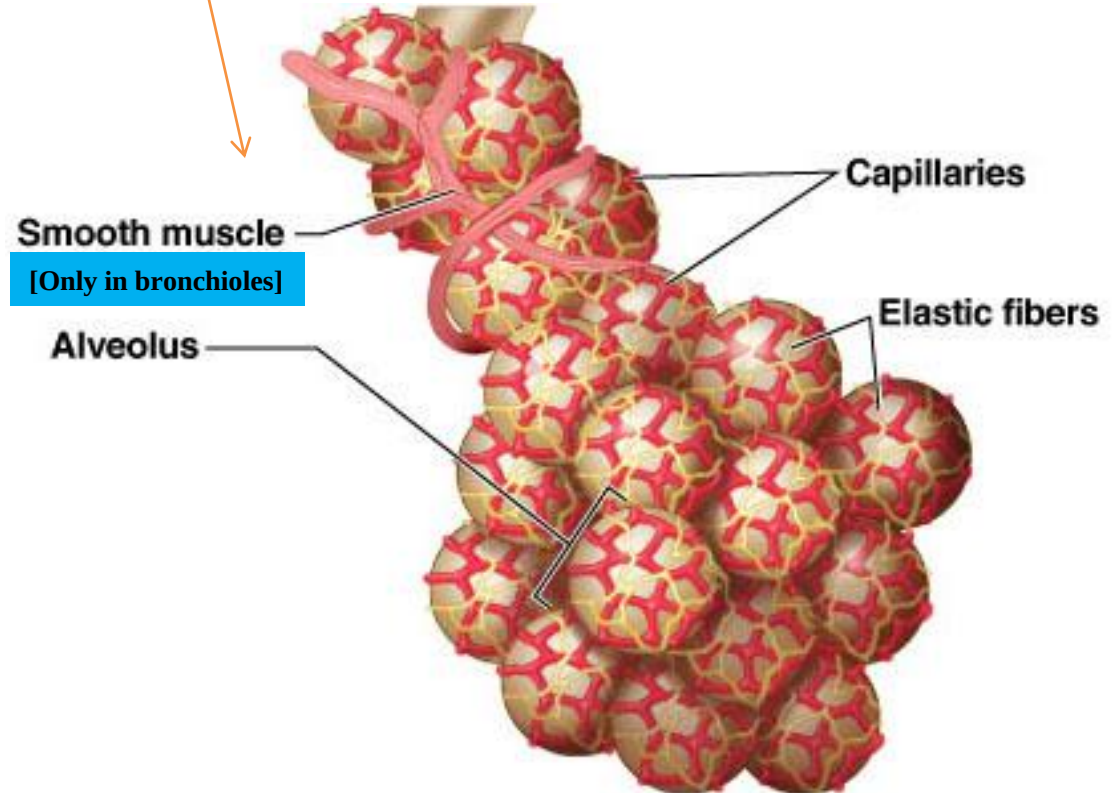
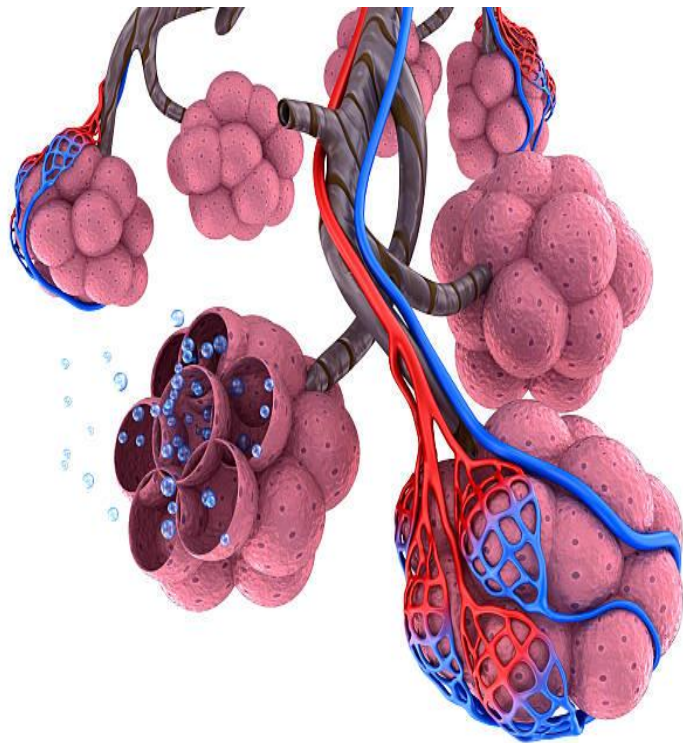
- ✓ The functional respiratory unit of the lung where gases are exchanged.
- ✓ A one cell layer thick & no muscle around the alveoli
- ✓ **Alveoli** are small outpouchings Or tiny air sacs [0.2 to 0.5mm in diameter] & are final branching of the respiratory tree & serve as a **primary sites of exchange of O₂ & CO₂**.
- ✓ Normal human lungs [2] have 300 million alveoli ➡ which provide **about 70 m² surface area for gas exchange.**

Functions of Major...

❑ Each alveolus is surrounded by blood capillaries supplied by the pulmonary artery.

✓ Capillaries covered Or surrounded ➡ 90% of the alveolar surface

❑ **Respiratory membrane** ➡ it is the barrier b/n the alveolar air and capillary blood.



Functions of Major...

The alveoli wall → lined by 3 types of Epithelial cells

1. Type I pneumocytes (**Type I cells**)
 - ✓ Account approximately 95% of the alveolar epithelial surface area
 - ✓ Used for gas diffusion b/n the alveolus and bloodstream.
2. Type II pneumocytes (**Type II cells**)
 - ✓ Are responsible for the production of pulmonary surfactants (detergent-like lipoprotein) ➡➡ prevent **alveolar collapse**.
 - ✓ Make up approximately 4% of the surface area
3. Type III pneumocytes (**Type III cells**)
 - ✓ Have alveolar phagocytic macrophage cells [found in alveolar cavities]
 - ✓ They defense against inhaled matter
 - ✓ Keep alveolar surfaces sterile by removing debris and microbes

Functions of Major...

Surfactant and its effect on surface tension

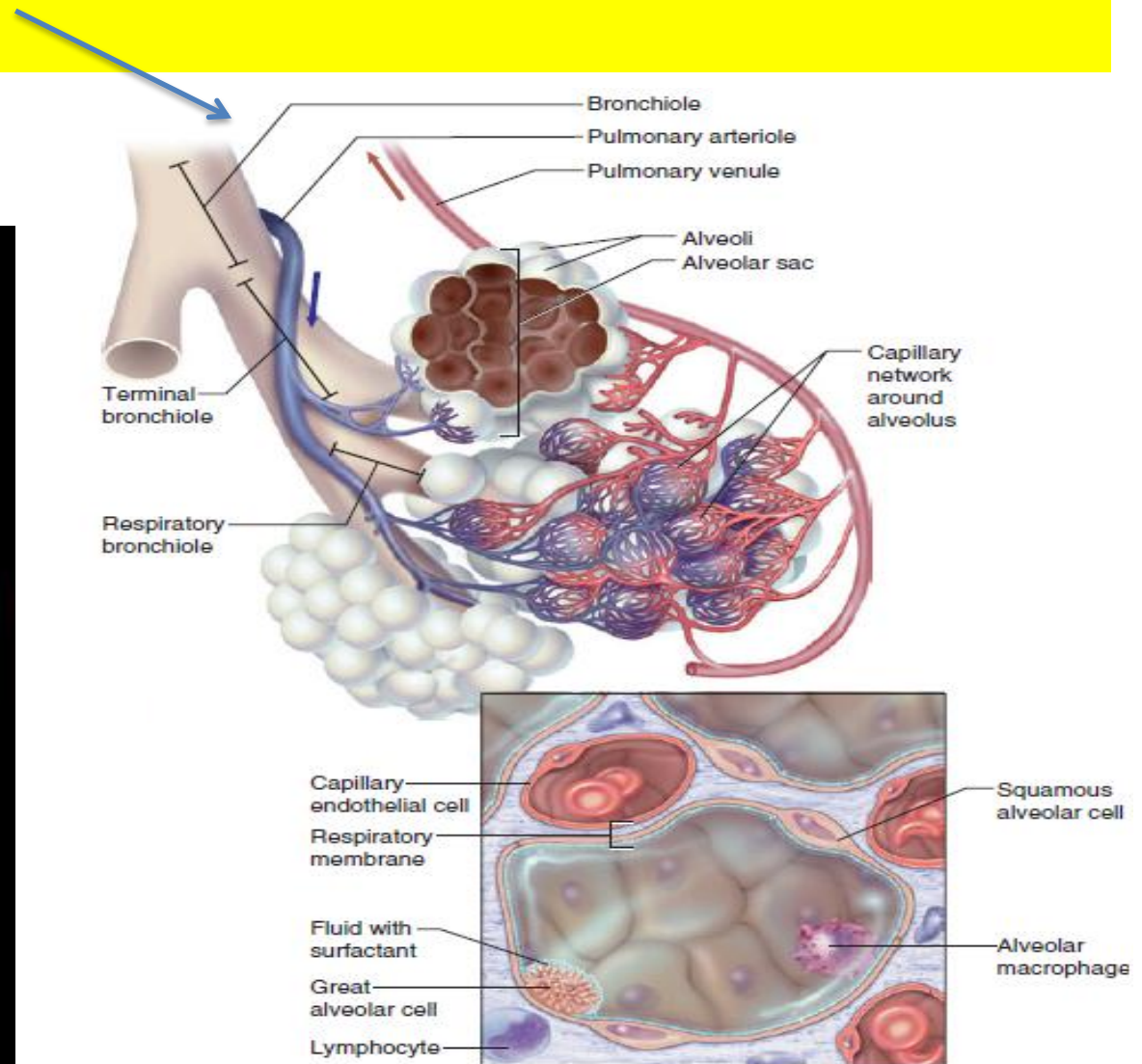
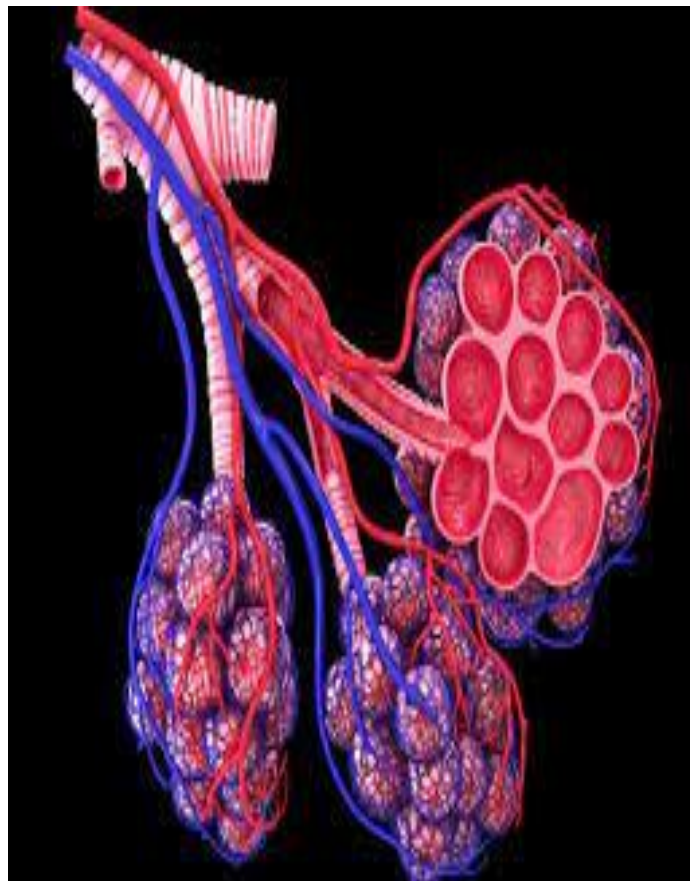
- ❑ **Surfactant** -A substance secreted by special epithelial cells called type II alveolar epithelial cells.
 - ❑ It greatly reduces the surface tension of water
- ❑ **Role of pulmonary Surfactant**
- ❑ Surfactant decreases surface tension which:
 - ✓ Increases pulmonary compliance
 - ✓ Reduce the effort needed to expand the lungs
 - ✓ Reduces tendency for alveoli to collapse (prevent lung collapse)
- ❑ Low secretion results respiratory distress syndrome in premature newborns
 - ✓ Lung collapse (atelectasis) due to reduced production of surfactant

Functions of Major...

Pulmonary Alveoli:

(a) Clusters of alveoli and their blood supply.

(b) Structure of an alveoli.



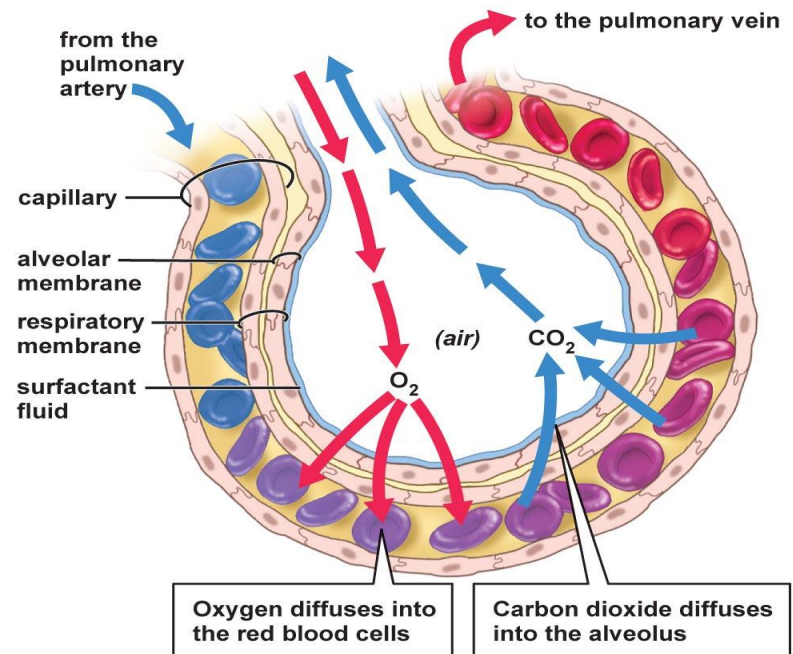
Capillaries surround Alveoli

Capillaries surround Alveoli

Functions of Major...

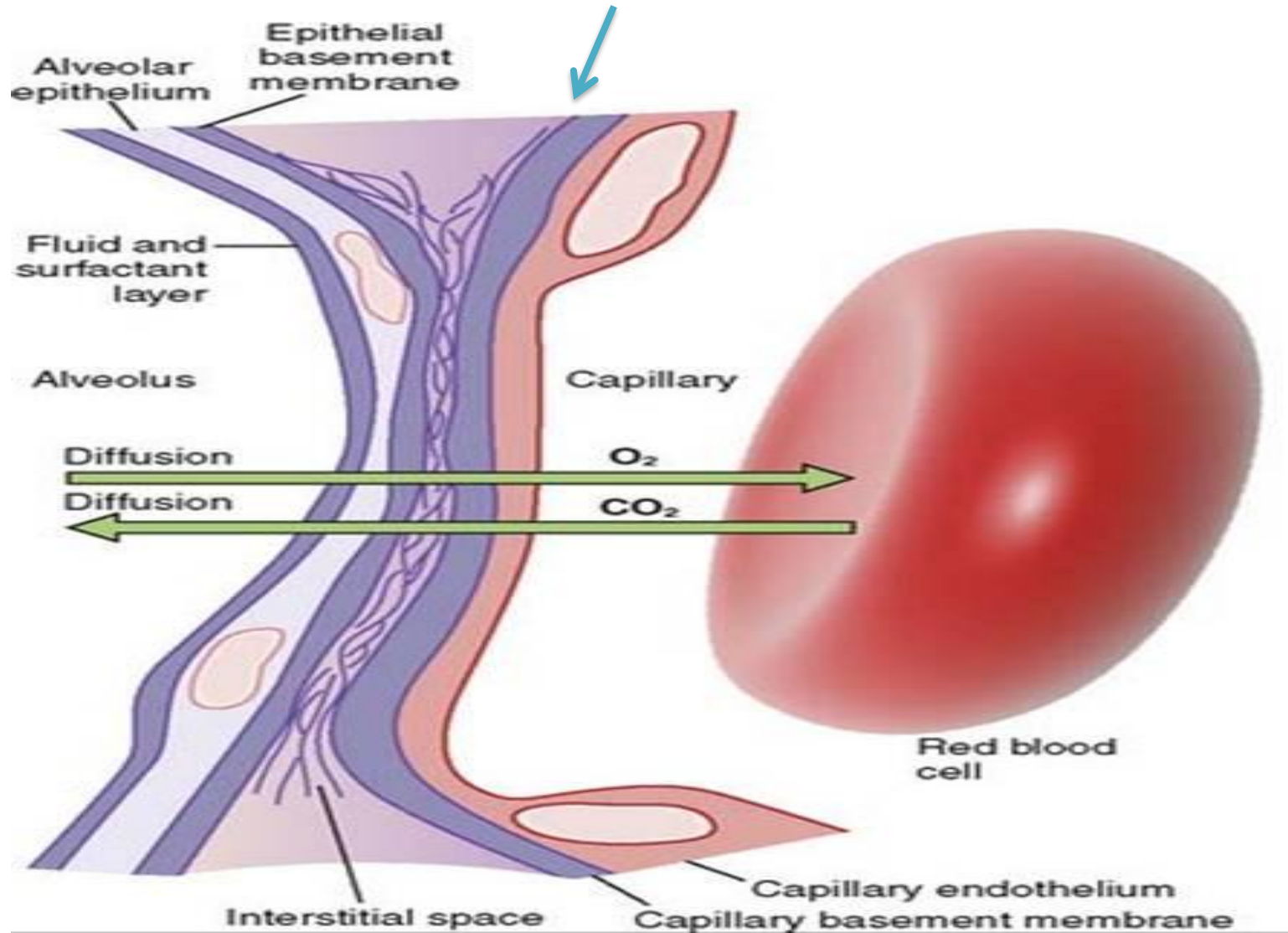
Respiratory Membrane [RM]

- ❑ The barrier b/n the alveolar air & pulmonary capillaries blood is called the **Respiratory membrane**.
- ❑ Gas exchange occurs across the **respiratory [Alveolo-Capillary] membrane** (0.5 μ m thick) ➡ allow rapid diffusion of gases b/n alveolus & capillary.
- ❑ **RM** consists of 4 layers:
 1. Type I & II cells of alveolus ➡
 2. Epithelial basement membrane ➡
 3. Capillary basement membrane ➡
 4. Capillary endothelium



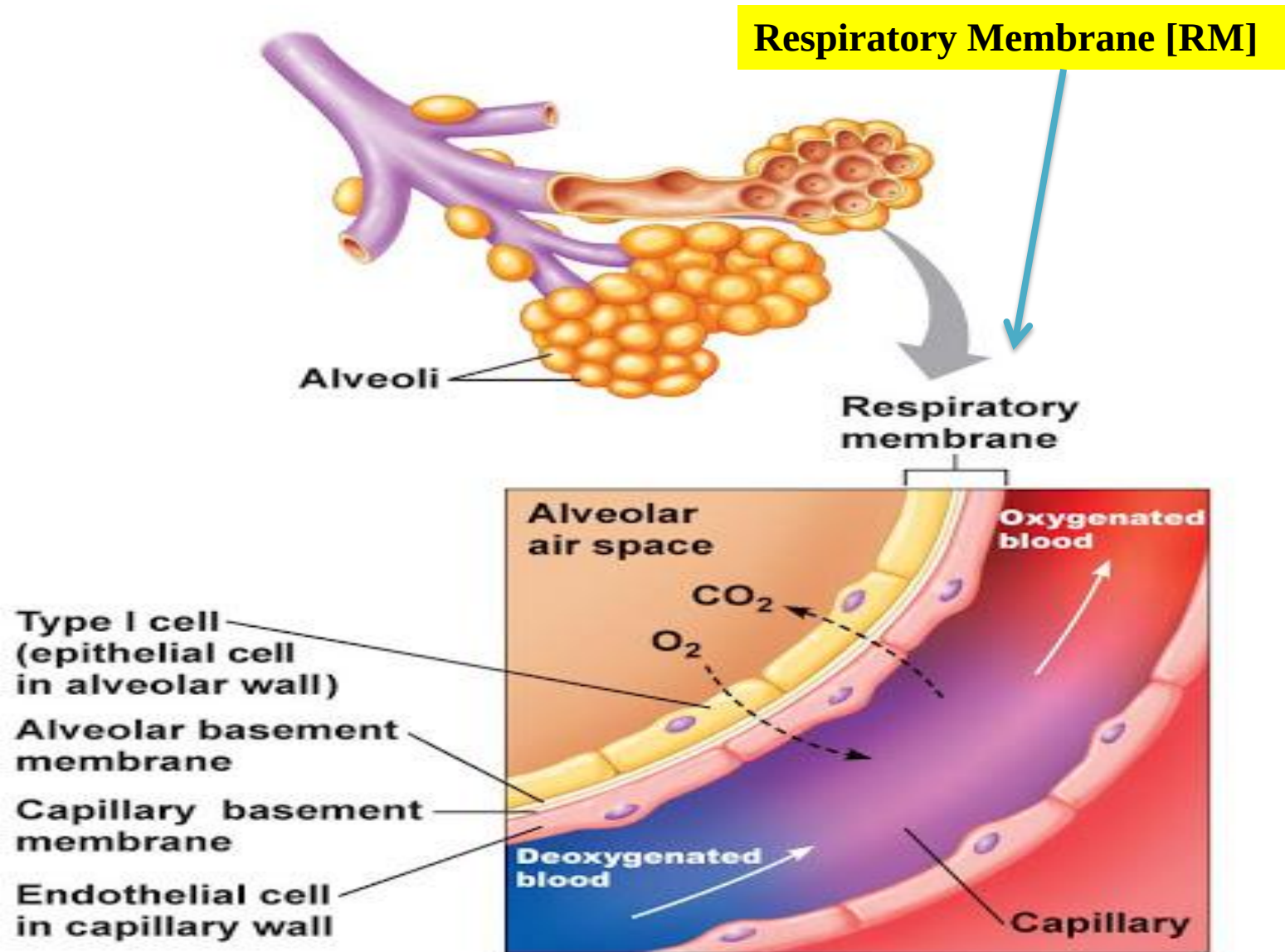
Functions of Major...

Respiratory Membrane [RM]



Alveolo-capillary membrane

Functions of Major...



Functions of Major...

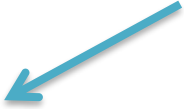
- ❑ **The Lung** ➡ It has Concave base ➡ & rest on diaphragm
 - ✓ **Right Lung** ➡ 3 lobes - superior, middle, inferior (accounts for 55% of the lung function)
 - ✓ **Left Lung** ➡ 2 lobes - upper and lower lobes. (accommodates heart) ➡ has less weight

The Pleura

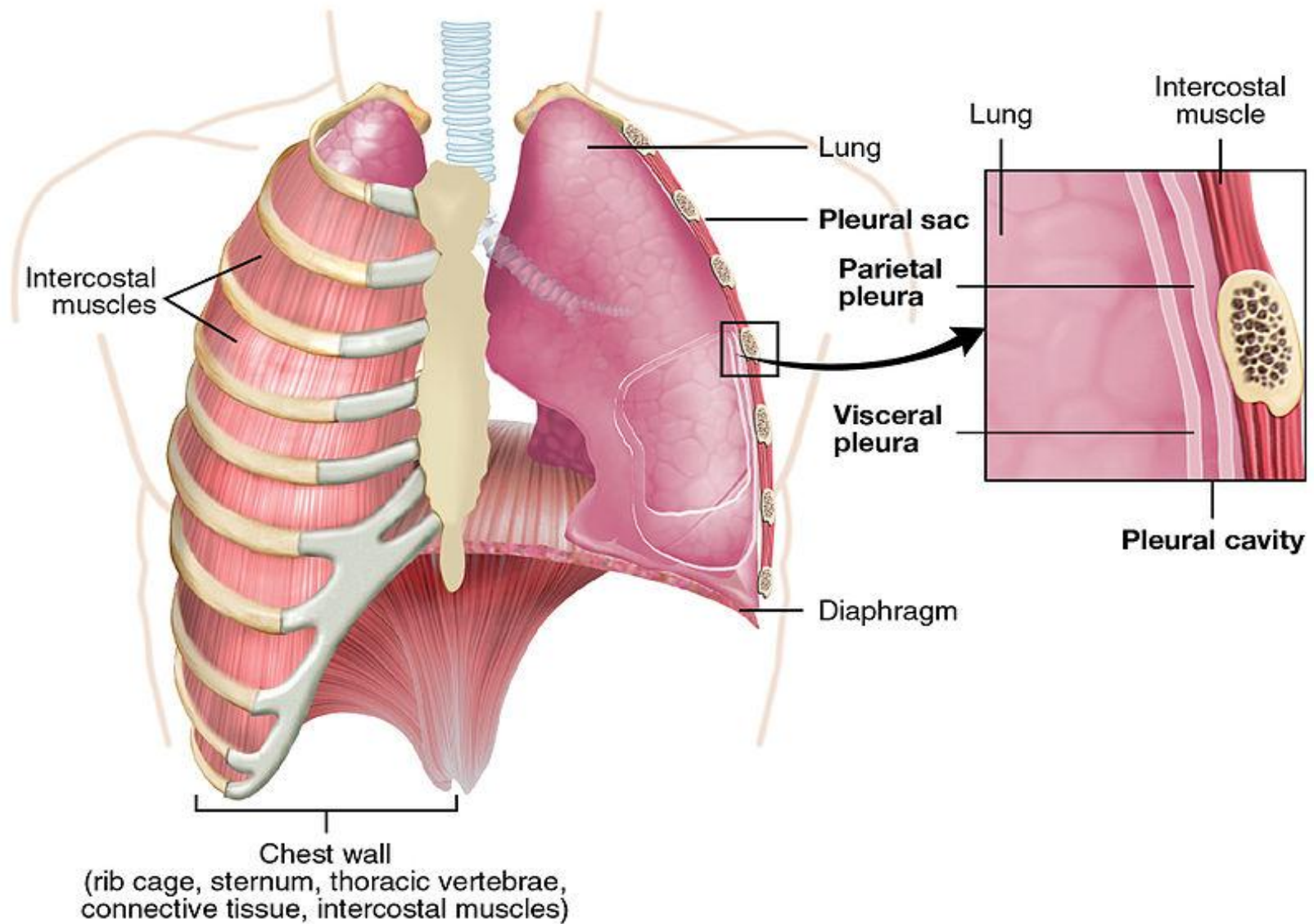
- ❑ In the thoracic cavity, each lung is surrounded by a **thin, moist, slippery pleural membrane**.
- ❑ The pleural membrane **has 2 layers**.
 1. The outer [parietal] layer lines **the inside of the rib cage & the diaphragm** while
 2. The inner [visceral] layer covers the lungs.
- ❑ **B/n the 2 layers** ➡ is the **intrapleural space [pleural cavity]**, which normally contains **fluid** secreted by the membranes ➡ **called pleural fluid**.

Functions of Major...

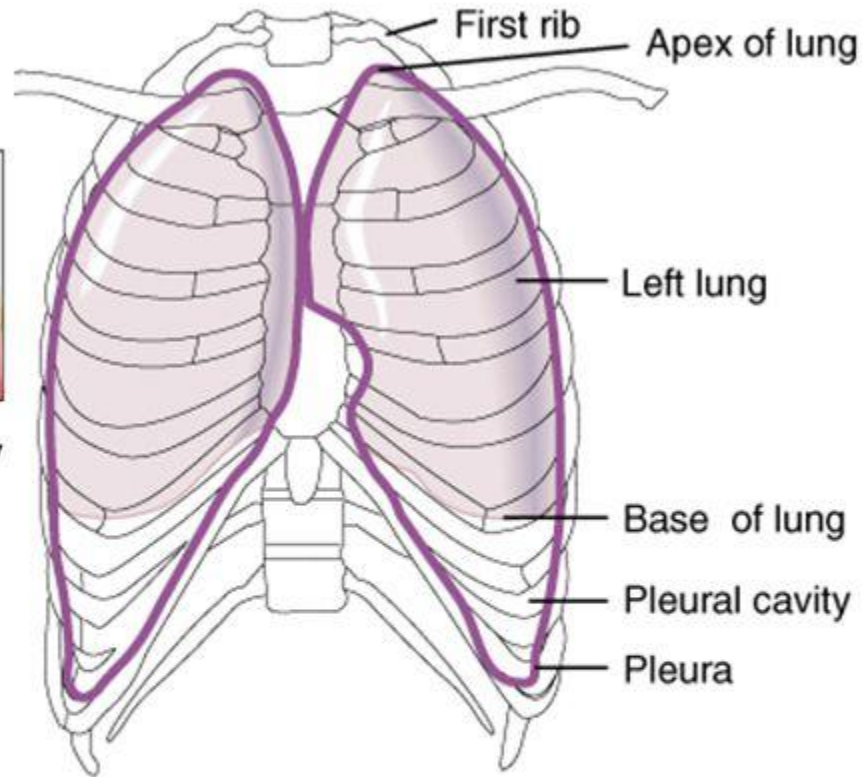
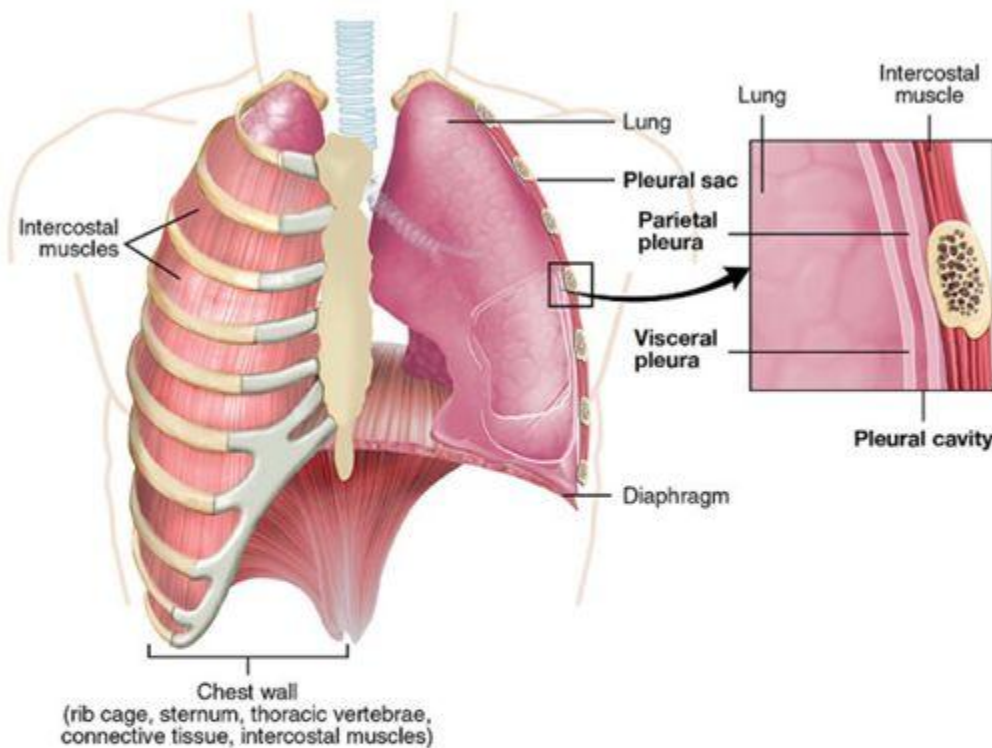
The pleural fluid performs 2 functions:

- ❑ It acts as a lubricant 
 - ✓ Allowing the visceral and parietal pleurae to slide past each other as the lungs and thorax change shape during respiration, &
- ❑ It helps hold the pleural membranes together.
 - ✓ The pleural fluid acts like a thin film of water b/n 2 sheets of glass (the visceral and parietal pleurae);
 - ✓ The glass sheets can slide over each other easily, but it is difficult to separate them.

Functions of Major...



Pleural Membrane & Pleural Cavity



- **Visceral pleura** covers lungs
- **Parietal pleura** lines ribcage & covers upper surface of diaphragm
- **Pleural cavity** is space between the two pleura
 - contains a small amount of fluid

Pressure Change & Airflow

❑ The flow of air into and out of the lungs is governed by 2 physical principles:

1. Changes in volume result in changes in pressure.

✓ As the volume of a container ↑, the pressure within the container ↓ &

✓ As the volume of a container ↓, the pressure within the container ↑.

❑ The muscles of respiration change thoracic volume [V] and therefore pressure [P] within the thoracic cavity.

Pressure Change & ...

2. Air flows from areas of higher to lower pressure [P]

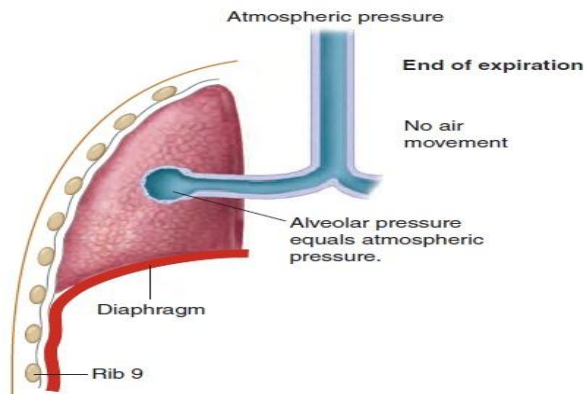
- ✓ If the P is higher at one end of a tube than at the other, air flows from the area of higher P toward the area of lower P.
- ✓ Air flows through the respiratory passages because of P differences b/n the outside of the body and the alveoli inside the body.
- ❑ These P differences are produced by changes in thoracic volume.

Pressure Change &...

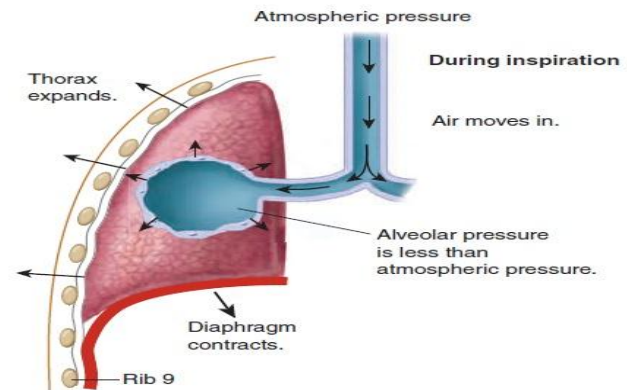
1. At the end of expiration,

- ✓ Alveolar pressure (the air pressure within the alveoli), is equal to atmospheric pressure (the air pressure outside the body).
- ✓ There is **no movement of air into or out of the lungs**
 - ✓ Because alveolar pressure and atmospheric pressure are equal.

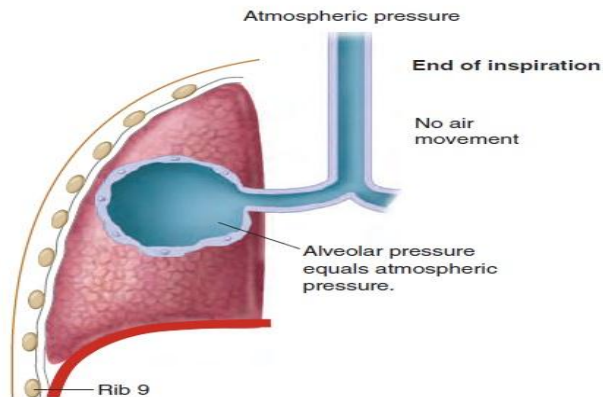
Pressure Change &...



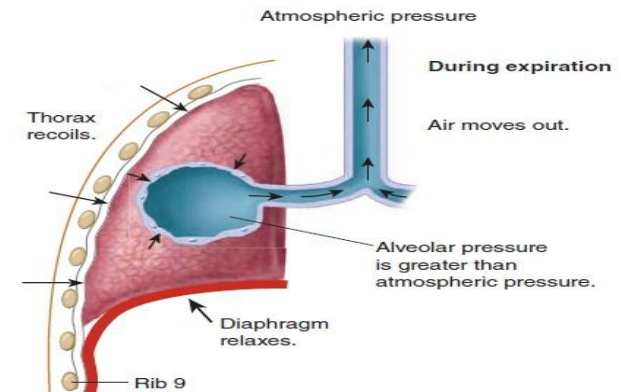
- 1 At the end of expiration, alveolar pressure is equal to atmospheric pressure, and there is no air movement.



- 2 During inspiration, increased thoracic volume results in increased alveolar volume and decreased alveolar pressure. Atmospheric pressure is greater than alveolar pressure, and air moves into the lungs.



- 3 At the end of inspiration, alveolar pressure is equal to atmospheric pressure, and there is no air movement.



- 4 During expiration, decreased thoracic volume results in decreased alveolar volume and increased alveolar pressure. Alveolar pressure is greater than atmospheric pressure, and air moves out of the lungs.

Pressure Change &...

2. During inspiration,

- ✓ Contraction of the **muscles of inspiration** ↑es ,the volume of the thoracic cavity.
- ✓ The **↑ed thoracic volume** causes the lungs to expand, resulting in an **↑ in alveolar volume**.
- ✓ As the alveolar volume ↑es, **alveolar pressure becomes less than atmospheric pressure**, &
- ✓ Air flows from the atmosphere to the alveoli

3. At the end of inspiration,

- ✓ The thorax and alveoli stop expanding.
- ✓ Alveolar pressure and atmospheric pressure become equal, and airflow stops .

Pressure Change &...

4. During expiration,

- ✓ The **thoracic volume** ↓es, producing a ↓ **in alveolar volume**.
- ✓ Consequently, alveolar pressure ↑es above the air pressure outside the body, and air flows from the alveoli to the outside .
- ✓ As expiration ends, the decrease in thoracic volume stops.

Physical Factors Influencing Ventilation [Airflow]

1. Lung Recoil

❑ During quiet expiration, thoracic and lung volume decrease because of **passive recoil** of the thoracic wall and lungs.

❑ **The recoil of the lungs occurs for 2 reasons:**

❑ **The elastic fibers** in the connective tissue of the lungs and

❑ **Surface tension** of the film of fluid that lines the alveoli.

Physical Factors Influencing...

2. Surface Tension

- ✓ Is the **attraction of liquid molecules** to one another inside the alveoli.
- ✓ This liquid coating [lining] the alveolar surface is **always acting to reduce the alveoli to the smallest possible size.**

3. Compliance

- ✓ The ability of the lungs to **stretch during inspiration**
- ✓ Lungs can stretch when under tension.

4. Surfactant

- ✓ A detergent-like complex secreted by Type II alveolar cells, **reduces surface tension** and helps keep the alveoli from collapsing.

Respiration

- ❑ It is the exchange of gases b/n atmosphere (atm) and body or
- ❑ The act or process of inhaling and exhaling of air
- ❑ **Respiration depends on:** air passages, lungs, pleural sac, thoracic cage, muscles of breathing, and nerve centers in the brain stem.
- ❑ **Functions of the respiratory system (RS)**
 - A. Respiratory functions (primary function) &
 - B. Non-respiratory functions

Cont'd

A. Respiratory function →→ For maintaining appropriate gas exchange

- ❑ To obtain oxygen for body's cells activities &

- ❑ To eliminate CO₂ that the cells produce

 - ✓ It is the primary function of the RS.

- ❑ **Such respiratory functions** consists of 4 distinct processes [stages]

1. Pulmonary ventilation

 - ✓ Inflow and outflow of air b/n Atm & lung alveoli

2. Alveolar ventilation [**External Respiration**]

 - ✓ Diffusion of O₂ & CO₂ b/n alveoli & pulmonary capillary blood

3. Transport of such gases in the blood

 - ✓ Transport of O₂ & CO₂ **b/n the lungs and tissues**

4. **Internal Respiration** [Exchange in tissue bed] → → → b/n systemic capillaries & cells

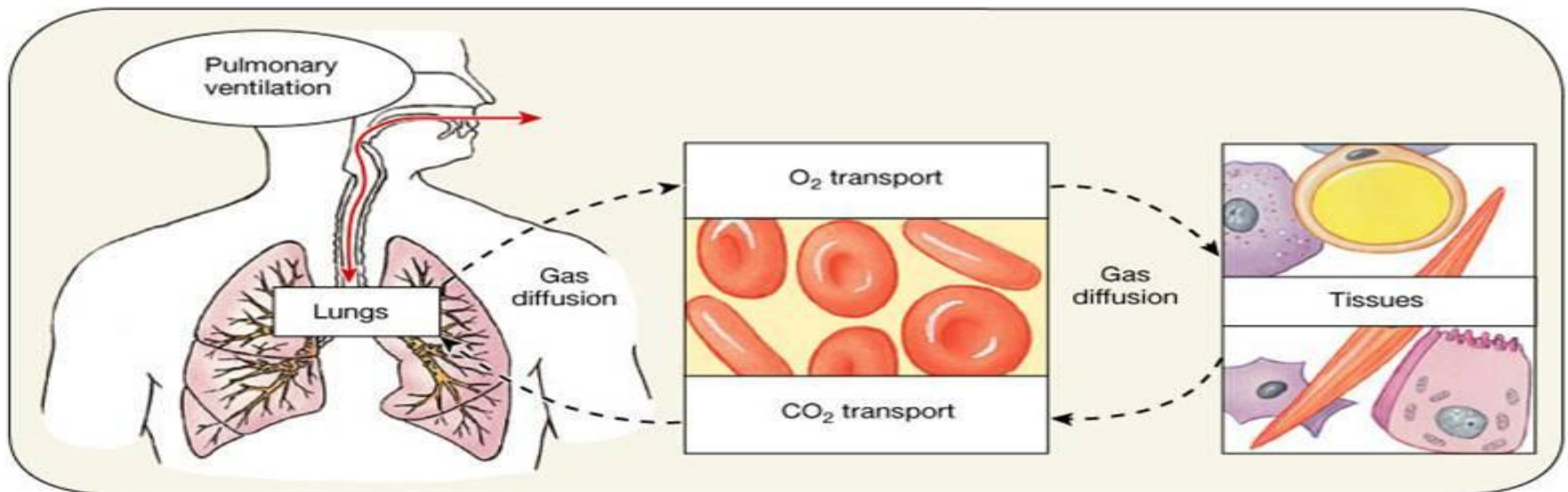
 - ✓ Diffusion of gases b/n the blood of the systemic capillaries and cells.

 - ✓ Cells require O₂ for cellular metabolism, growth & division

Cont'd

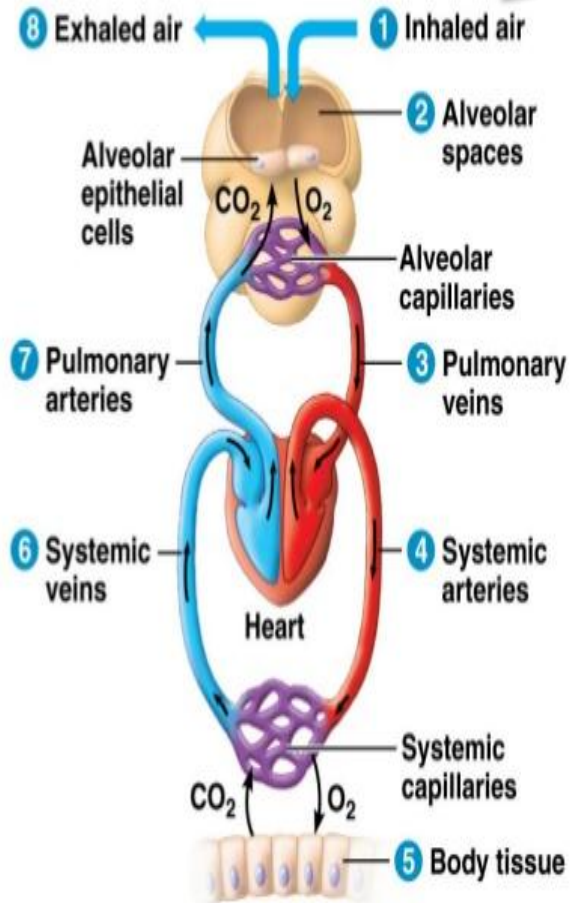
Stages of Respiration

1. **Pulmonary ventilation** ➡ gas exchange b/n the atmosphere and lungs
2. **External respiration** ➡ gas exchange b/n the lungs and blood (O_2 loading and CO_2 unloading).
3. **Transport of respiratory gases** ➡ via movement of blood O_2 from the lungs to the cell and tissues.
4. **Internal respiration** ➡ gas exchange b/n the capillaries and the tissues (O_2 unloading and CO_2 loading).

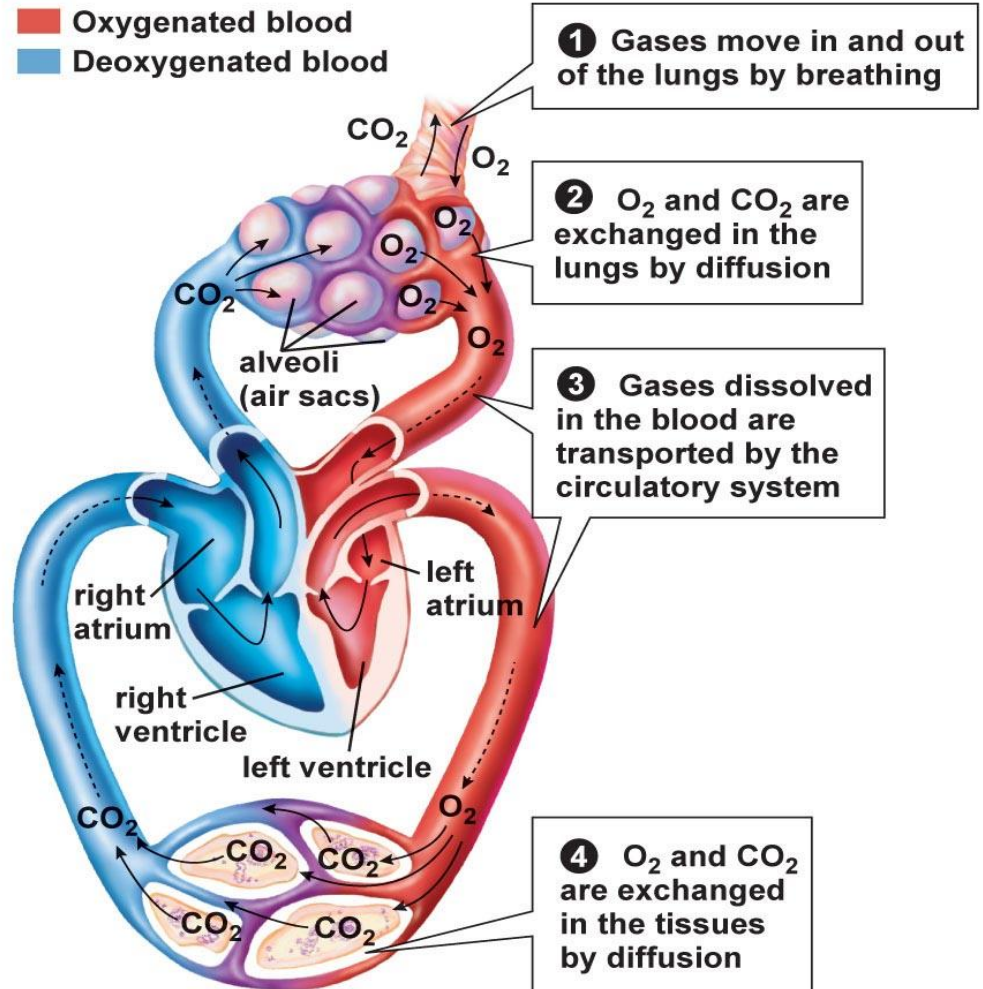


Cont'd

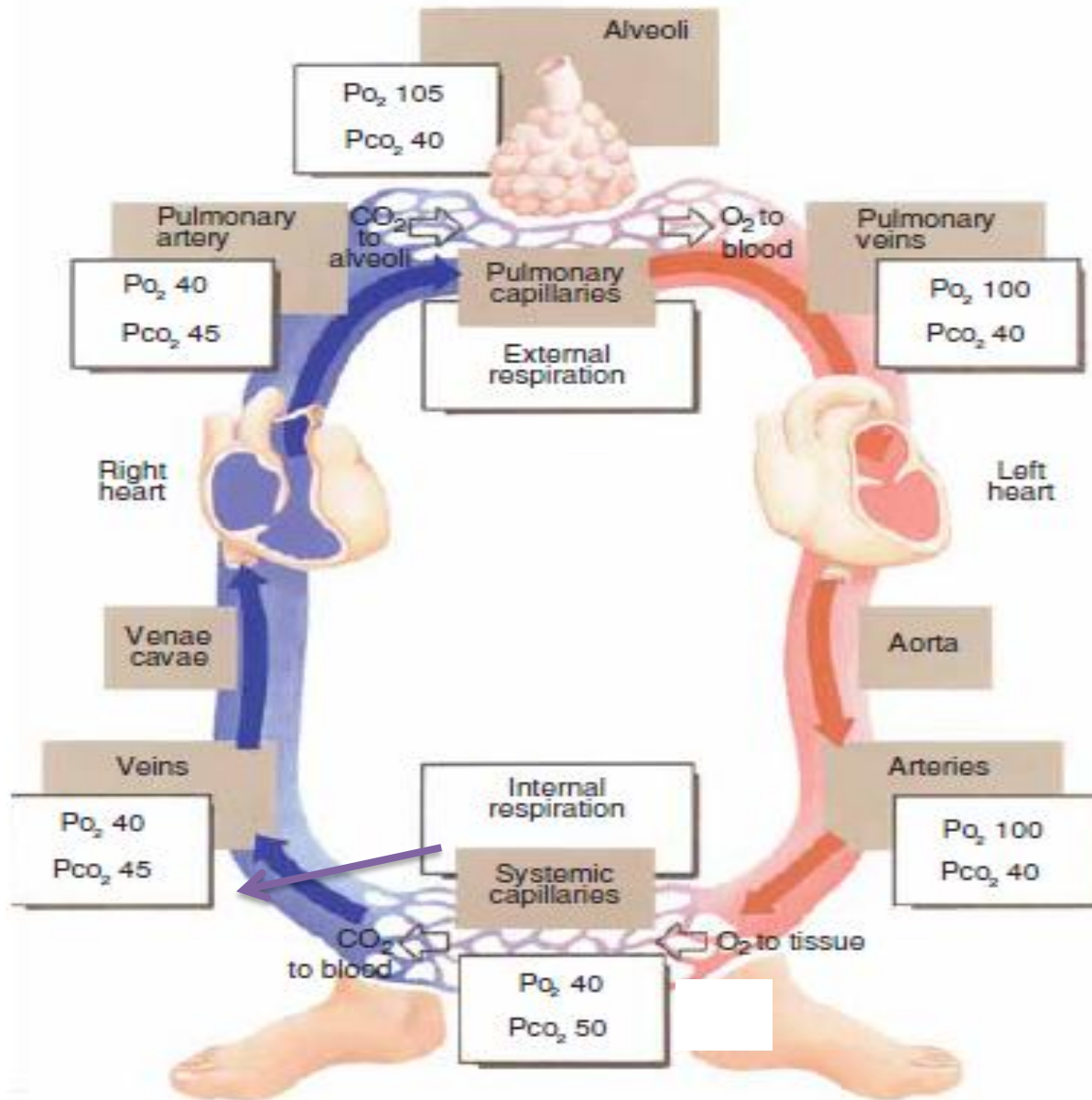
Circulation and Gas Exchange



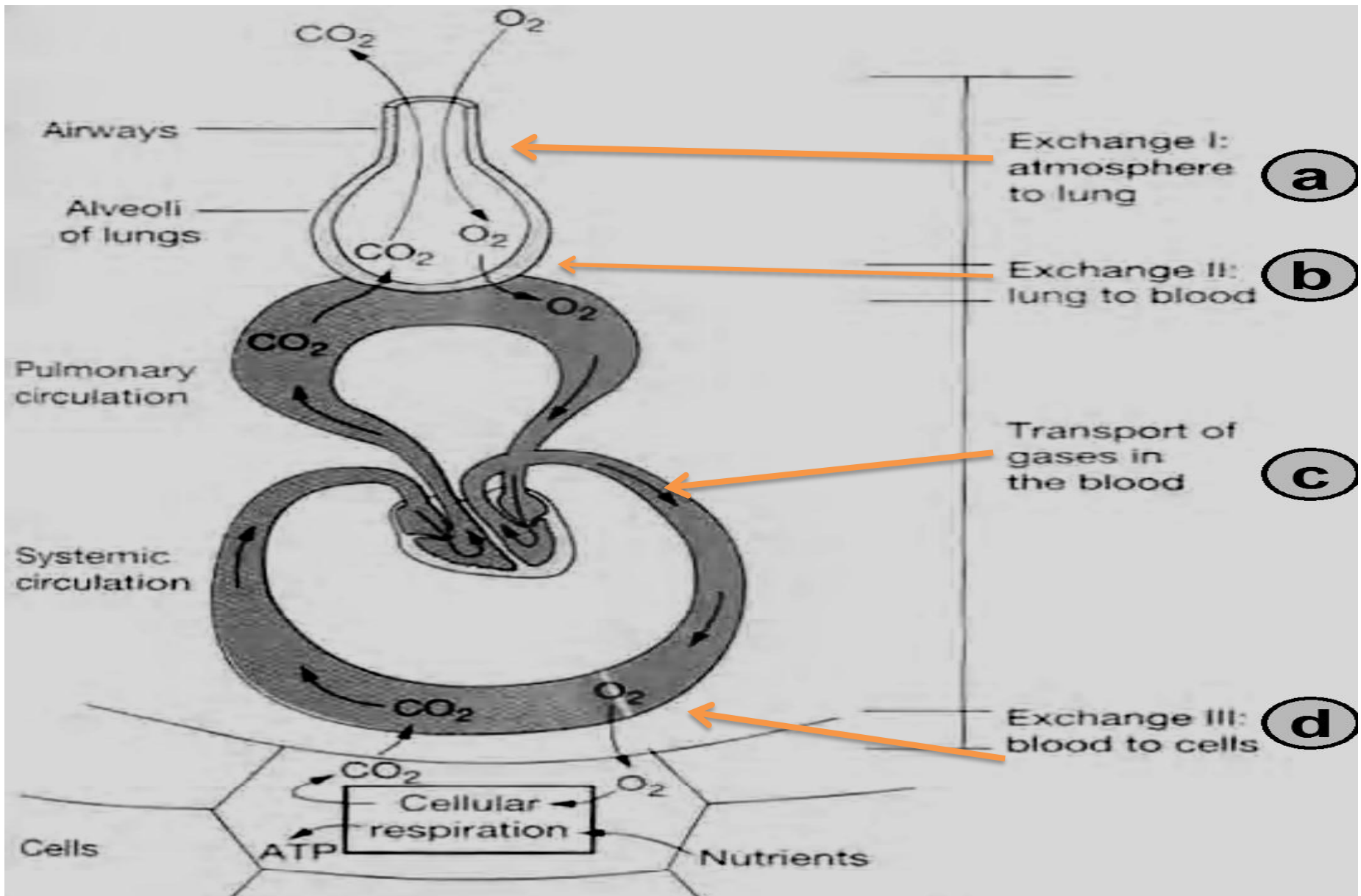
The path of respiratory gases in the circulatory system



Cont'd



Cont'd



In summary

Phases of respiration ➡ generalized into 2

1. External respiration (I & II)

- ✓ Inhalation of gas (O_2) from Atm into → alveoli → & into pulmonary capillary beds &
- ✓ Exhalation of CO_2 from pulmonary capillary to → alveoli → & into the Atm.

☐ Includes:

☐ **Pulmonary ventilation = exchange I**

- ✓ B/n Atm and lung (alveoli)

☐ **Alveolar ventilation = exchange II**

- ✓ B/n lung (alveoli) and pulmonary blood (pulmonary capillary)

2. Internal (cellular) respiration (III)-b/n systemic capillaries & cells

- ✓ O_2 is utilized for metabolic activities &
- ✓ CO_2 produced as a result of metabolic activities of cells

Cont'd

B. Non-respiratory functions of RS

1. A means for water & heat loss (rid of excess water & heat) & deglutition
2. It enables vocalization: speech, singing,
3. Defense against foreign matter & noxious stimuli like inhaled particles,
 - ✓ Eg. by alveolar macrophages
4. Regulation of various humoral concentrations through the endothelium, e.g. Ang.I to Ang.II by ACE.
 - ✓ The lungs convert angiotensin I $\Rightarrow$ angiotensin II
 - ✓ By producing angiotensin converting enzyme = ACE
 - ✓ ACE is produced in pulmonary capillary
5. Olfaction
 - ✓ The nose serves as organ of smell
 - ✓ Olfactory receptors are in the mucus membrane lining the upper nostril
6. Maintenance of normal - acid base balance $\rightarrow$ regulate blood P^H
 - ✓ By altering the amount of CO_2 [generating H^+]

The Respiratory Muscles

❑ The muscles of breathing are divided into 2 main groups:

1. Muscles of Inspiration
2. Muscles of Expiration

1. Inspiratory Muscles:

❑ The muscles of inspiration are principal muscles and accessory muscles

❑ Principal inspiratory muscles work at resting respiration are:

- ✓ The diaphragm [70%] and
- ✓ External intercostal muscles [30%]

The Respiratory Muscles...

❑ The accessory muscles of inspiration assists the main muscles of inspiration to produce forced inspiration.

❑ These muscles include:

- ✓ The sternomastoids [sternocleidomastoid]
- ✓ The scaleni
- ✓ The serratus posterior inferior
- ✓ The serratus posterior superior

The Respiratory Muscles...

2. Expiratory Muscles:

- ❑ Normal resting expiration is a passive process produced by elastic recoil of the lungs
- ❑ Muscles of expiration are only **accessory muscles** w/c work to produce forced expiration are:
 - ✓ Internal IC muscles
 - ✓ The abdominal wall muscles including the pelvic floor muscles

Mechanics of Pulmonary Ventilation

- ❑ Involves inspiration and expiration
- ✓ **Work under the principle of Boyle's law** : $P \propto 1/V$, meaning $\uparrow P \Rightarrow \downarrow V$
 - ✓ Gases diffuse from area of high 'P' to area of low 'P' (from low to high V)

A. Inspiration:

❑ The most fundamental muscles of inspiration:

1. External intercostal muscles &
2. Diaphragm
3. The accessory muscles, contract during inspiration:
 - a. Sternocleidomastoid muscles - lift the sternum upward
 - b. Anterior serrati - lifts many of the ribs
 - c. The scalene - lifts the first two ribs

Mechanics of Pulmonary Ventilation...

❑ Inspiration involves contraction of inspiratory muscles:

1. Contraction of external intercostal muscles



❑ Elevation of ribs & sternum

- ✓ Ribs and sternum move outward & upward



❑ Increased front- to-back dimension of thoracic cavity (↑volume)



❑ Lowers air pressure in lungs → air moves from atm into lungs.

2. Contraction of diaphragm → diaphragm moves downward

- ✓ Increases vertical dimension of thoracic cavity (chest cavity)



- ✓ Lowers air pressure in lungs → air moves into lungs

❑ There is also contraction of accessory inspiratory muscles

- ✓ Enhance the effects of fundamental inspiratory muscles

Mechanics of Pulmonary Ventilation...

B.Expiration:

- ❑ Expiratory muscles are abdominal muscles & internal intercostal muscles

1.Abdominal muscles - the most important muscles of expiration



- ❑ Their contraction push the abdominal contents upward against bottom of diaphragm → diaphragm relax (move upward)



- ❑ ↓ the volume of thoracic cavity (↑ pressure in lungs)



Air is exhaled

2.Contraction of internal intercostal muscles



- ❑ Return ribs, & sternum to resting position



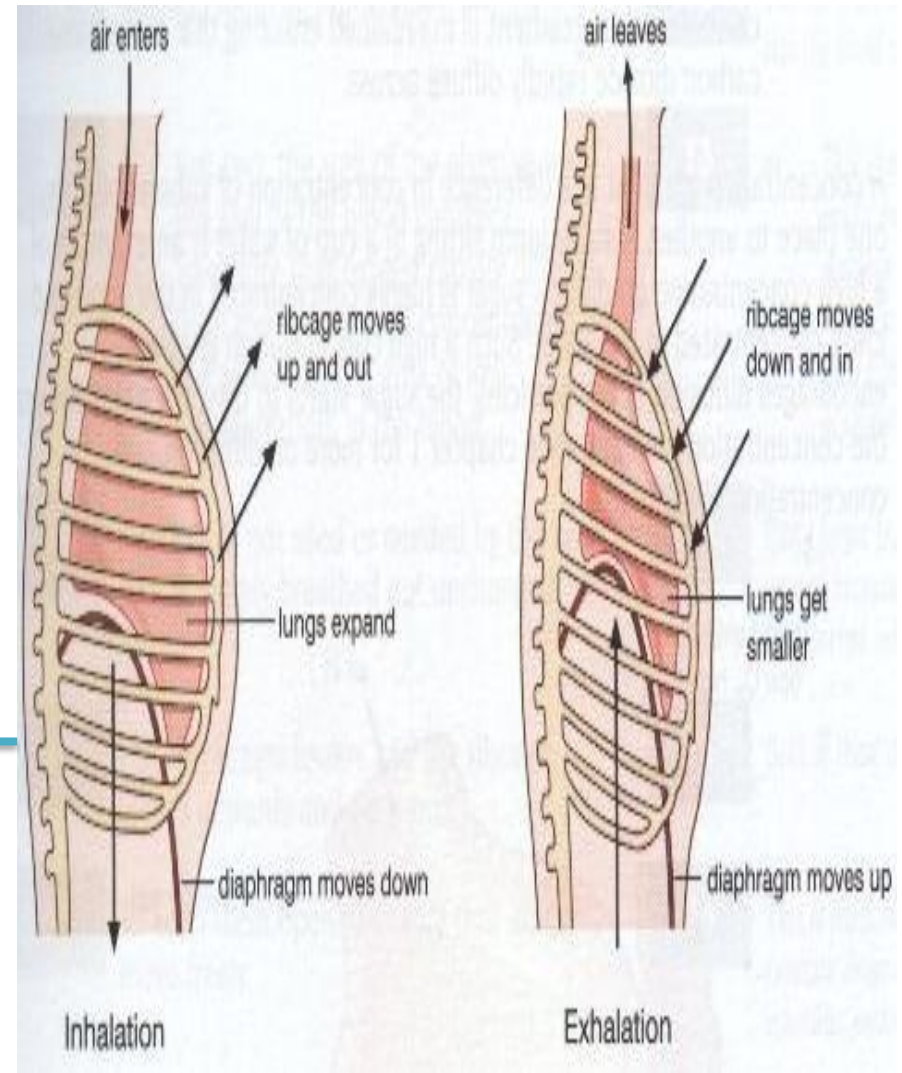
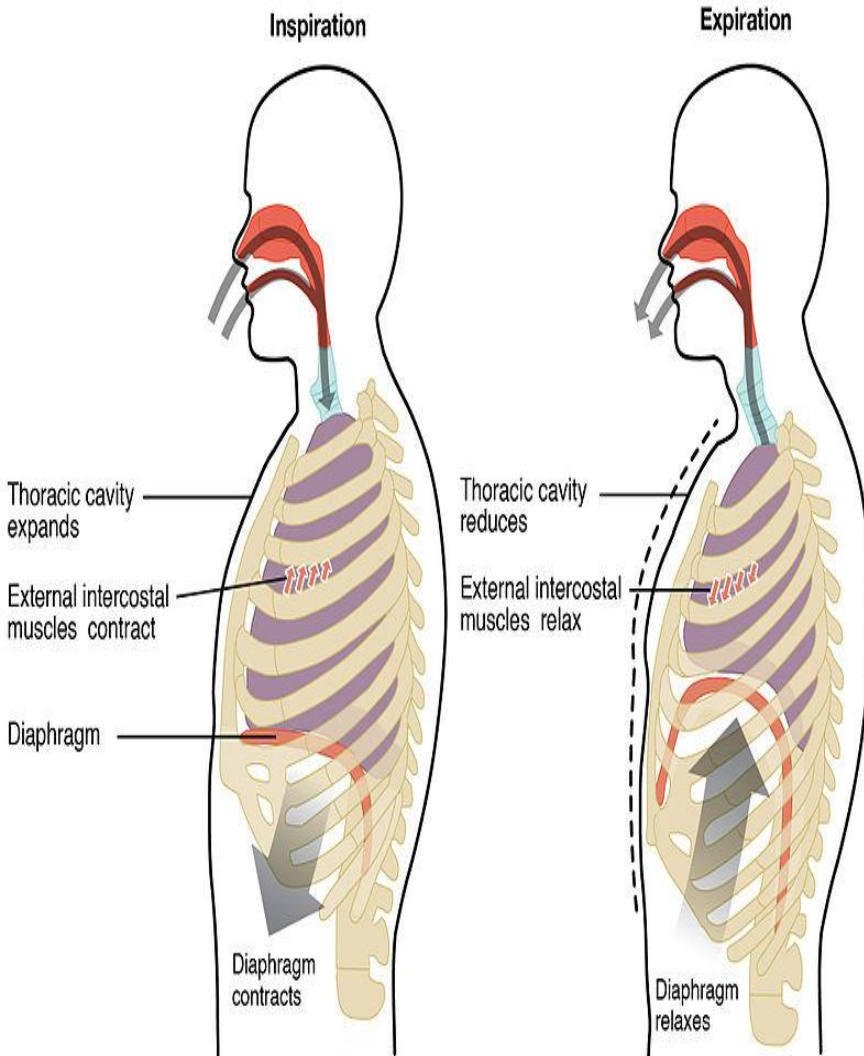
- ❑ Restores thoracic cavity to pre-inspiratory volume

→ ↑ pressure in lungs



Air is exhaled

Mechanics of Pulmonary Ventilation...



In summary

- ❑ Pulmonary Ventilation is ➡ the movement of air into/out of alveoli
- ❑ Air flows from area of high pressure to low
- ❑ 3 type of pressures are important in ventilation
 1. **Atmospheric (barometric) pressure** ➡ 760 mm Hg at sea level
 2. **Intra-alveolar pressure (intrapulmonary pressure)** ➡ the pressure within the alveoli = 760 mmHg
 3. **Intrapleural pressure (intrathoracic pressure)** ➡ the pressure within the pleural sac = 756 mmHg

Cont'd

Alveolar ,Atm and Pleural pressure changes in the Respiratory Cycle

1. During inspiration ➡ alveolar pressure $<$ atm pressure
2. During expiration ➡ alveolar pressure $>$ atm pressure
3. Through out the respiratory cycle
 - ✓ The pleural pressure $<$ alveolar pressure $\Rightarrow$ to prevent lung collapse
 - ✓ If pleural p $>$ atm P $\Rightarrow$ the lung will collapses
 - ✓ B/c of the pressure exerted on the exterior surface of the lung

Lung Volumes and Capacities

- ❑ On average, in healthy young adults, the maximum air that the lungs can hold is about 5.8 liters.
 1. Age
 2. The distensibility of the lungs, and
 3. The presence or absence of respiratory disease affect this total lung capacity.
- ❑ The changes in lung volume that occur with different respiratory efforts can be measured using a **spirometer**.

Lung Volumes and Capacities...

□ The 4 standard Lung volumes are:

1. Tidal volume (TV)
2. Inspiratory reserve volume (IRV)
3. Expiratory reserve volume (ERV)
4. Residual volume (RV)

- ✓ **TV** ➡ volume of air entering or leaving the lungs during a single breath (500 ml)
- ✓ **IRV** ➡ extra volume of air inhaled after a normal inspiration (3000 ml).
- ✓ **ERV** ➡ extra volume of air that can be expelled after a normal expiration (1000 ml).
- ✓ **RV** ➡ volume of air remaining in the lungs after a maximal expiration (1200 ml).

Lung Volumes and Capacities...

☐↓ Lung Capacities:

- ☐ Consist of 2 or more lung volumes in combination
- ☐ The 5 standard lung capacities are:
 - ☐ Functional residual capacity (FRC)
 - ☐ Inspiratory capacity (IC)
 - ☐ Expiratory capacity (EC)
 - ☐ Vital capacity (VC)
 - ☐ Total lung capacity (TLC)

FRC: volume of air remaining in the lungs at the end of a normal expiration (2200 ml).
($FRC = ERV + RV$)

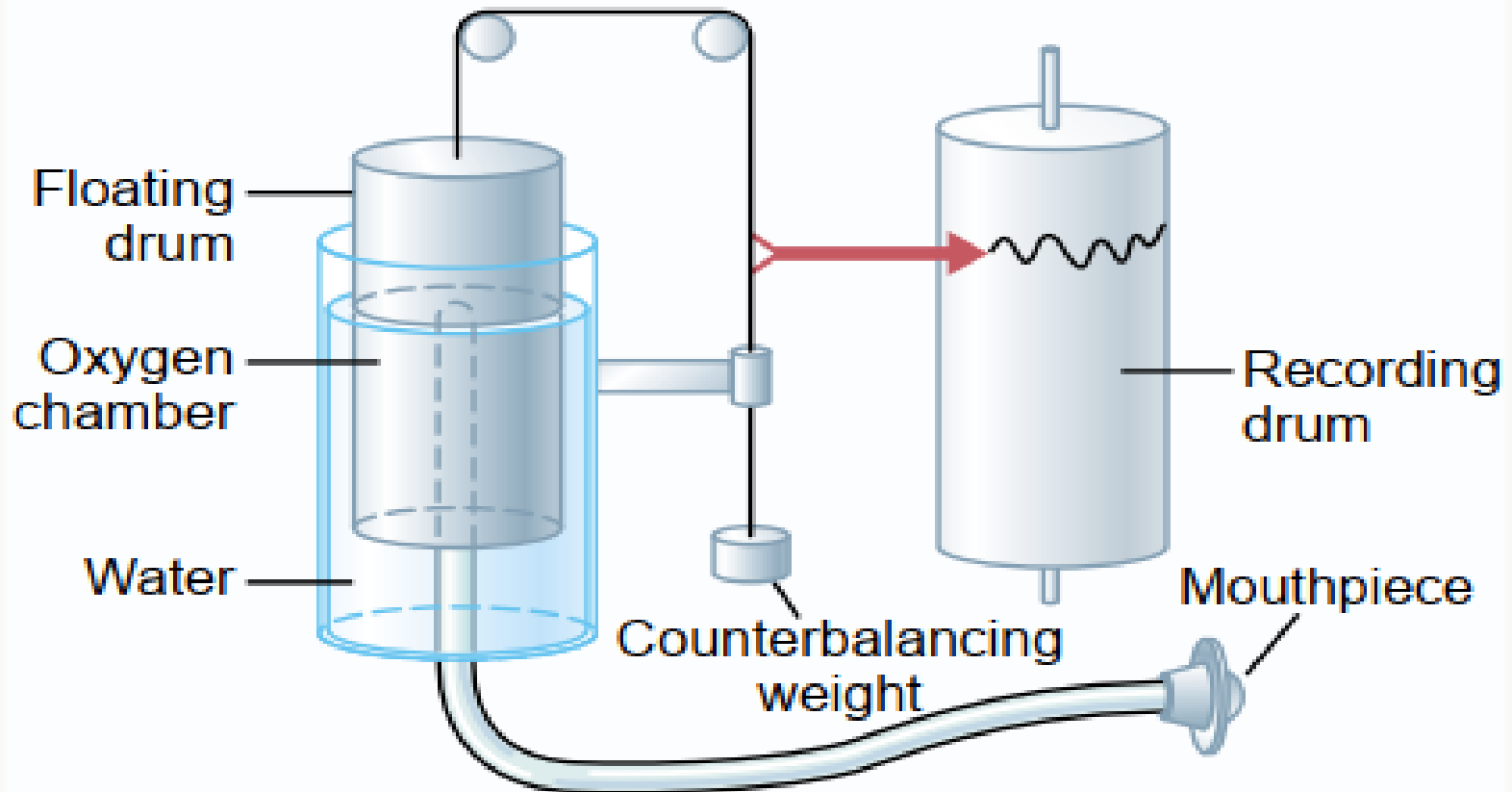
IC: Max. volume of air that can be inspired from FRC (3500ml)
($IC = IRV + TV$)

VC: max. volume of air that can be expired after maximal inspiration (4500 ml).
($VC = IRV + TV + ERV$)

TLC: volume of air in the lungs after a maximal inspiration (5700 ml).
($TLC = VC + RV$)

Spirometer

❑ Is a device that measures the volume of air breathed in and out



Spirometer...

- ❑ 2 general categories of **respiratory dysfunction** **[diseases]** yield abnormal results during spirometry.

1. **Obstructive lung disease** (e.g. asthma or bronchitis, emphysema).

- ✓ There is a difficulty in **moving gas out of the lung** due to high airway resistance
- ✓ FEV1/FVC far below 80%, where :-
 - ✓ Forced vital capacity (FVC)- is forced maximal expiration into a spirometer after maximal inspiration.
 - ✓ FEV1 is forced expiratory volume of air expired in the first second of expiration.

Spirometer...

2. Restrictive lung disease (e.g. pulmonary fibrosis)

- ✓ Replacement of delicate lung tissue with thick, fibrous tissue
- ✓ Lungs are **less compliant** (not expand easily) than normal, so it is difficult to move gas into the lungs

A. Pulmonary & Alveolar ventilation

A. Pulmonary Or Minute ventilation

- ✓ The volume of air breathed in and out in one minute.
- ✓ Pulmonary ventilation= Tidal volume x Respiratory Rate [RR=Bf]

$$(\text{ml/min}) = (\text{ml/breath}) (\text{breaths/min})$$

$$6000 \text{ ml/min} = 500 \text{ ml/breath} \times 12 \text{ breaths/min}$$

- ✓ These values apply to conditions of normal, quiet breathing
- ✓ TV and RR increase substantially during exercise

Dead space

- ❑ **Dead space** is the volume of air that is inhaled and does not take part in the gas exchange, because it either remains in the conducting airways or reaches alveoli that are not perfused or poorly perfused.
- ✓ About 150 ml of it fills the conducting division of the airway.
- ✓ Since this air cannot exchange gases with the blood, it is called **dead air**.
- ❑ **Anatomical dead space:**
 - ❑ Volume of air in **conducting airways**
 - ❑ This volume is considered anatomic dead space b/c air within these airways is **useless for exchange**.
 - ❑ NB: Conducting airways do not contain alveoli to participate in gas exchange.

Dead space...

❑ Alveolar dead space:

- ✓ Is the volume of air that enters unperfused alveoli.
- ✓ In other words, these alveoli receive airflow but **no blood** flow; with no blood flow to the alveoli, gas exchange cannot take place.

Physiological dead space:

- ✓ Anatomical dead space + alveolar dead space
- ✓ Determined by measuring the amount of CO₂ in the expired air.

NB: Only 350 ml are exchanged b/n the **atmosphere and alveoli** because of the 150 ml occupying the anatomic dead space.

B. Alveolar ventilation

- ❑ The volume of air **exchanged** between the atmosphere and alveoli per minute
- ❑ Alveolar ventilation is less than pulmonary ventilation because of **dead space**
- ❑ Alveolar ventilation = (tidal volume – dead space volume)

x respiratory rate

- ❑ Alveolar ventilation = (500 ml/breath -150 ml dead space
volume) x12 breaths/min
= 4200 ml/min

- ❑ Thus, with quiet breathing, alveolar ventilation is 4200 ml/min,
whereas pulmonary ventilation is 6000 ml/min.

Summary of Spirometry

Pulmonary Volumes and Capacities

- ❑ **Spirometry** ➡ is the process or the procedure of measuring volumes of air that move into and out of the respiratory system
- ❑ **Spirometer** ➡ is the device that is used to measure these pulmonary volumes
- ❑ Measurements of the respiratory volumes can provide information about the health of the lungs.



Summary of Spirometry...

Lung volumes and capacities

- ❑ **Lung volume** - amount of air inspired or expired
- ❑ **Lung capacity** - Σ of two or more lung volumes
 - ✓ Measured by Spirometer (device)
 - ✓ The method is spirometry
 - ✓ The tracing is spirogram

There are 4 pulmonary lung volumes

- | | | |
|----|----------------------------------|----------|
| 1. | Tidal Volume (TV) | = 500ml |
| 2. | Inspiratory reserve volume (IRV) | = 3000ml |
| 3. | Expiratory reserve volume (ERV) | = 1100ml |
| 4. | Residual volume (RV) | = 1200ml |

5 types pulmonary Capacities

- | | | |
|----|----------------------------------|---------|
| 1. | The total lung capacity | = (TLC) |
| 2. | The functional residual capacity | = (FRC) |
| 3. | The vital capacity | =(VC) |
| 4. | The inspiratory capacity | =(IC) |
| 5. | The expiratory capacity | =(EC) |

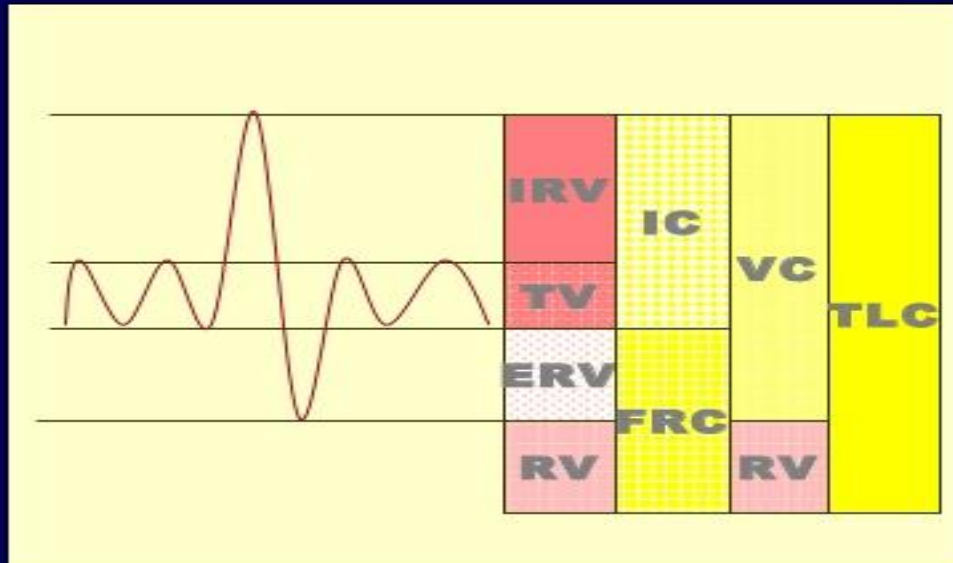
Summary of Spirometry...

❖ Summary [Lung volumes]

1. Tidal volume - the volume of air that is inspired and expired during quite breathing.

✓ Average about 500ml

Tidal Volume (TV)

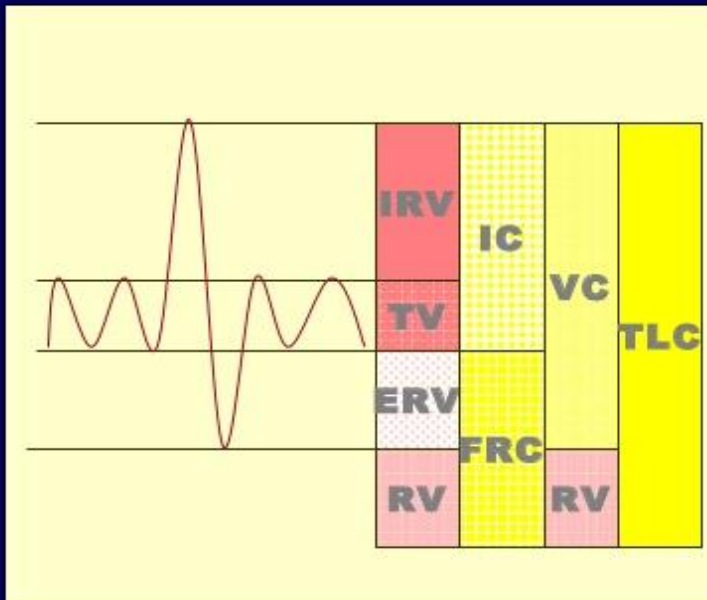


- Volume of air inspired and expired during normal quiet breathing

Summary of Spirometry...

- 2.**IRV**→ The maximum volume of air that can be inhaled during forced breathing.
✓ Averages about 3000 mL.

Inspiratory Reserve Volume (IRV)



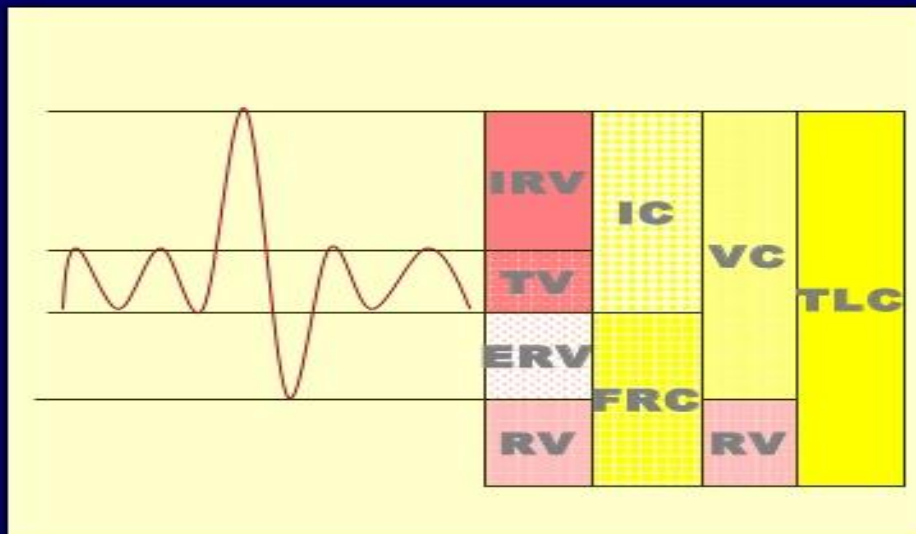
- The maximum amount of air that can be inhaled after a normal tidal volume inspiration

Summary of Spirometry...

3. **ERV** ➔ the maximum volume of air that can be exhaled during forced breathing.

✓ Averages about 1100 ml

Expiratory Reserve Volume (ERV)



- Maximum amount of air that can be exhaled from the resting expiratory level

4. **Residual volume (RV)** ➔ the volume of air remaining in the lung after forceful expiration.

✓ RV = 1200ml

Summary of Spirometry...

Disorders during air move in and out of the lungs

1. Obstructive lung disease (OLD)

- ☐ This is when air has trouble flowing out of the lungs due to resistance.
- ☐ This causes a decreased flow of air out (difficult to breath out)
- ☐ Reflects flow resistance property of respiratory airway
 - ✓ Asthma (bronchospasm)
 - ✓ COPD (which includes chronic bronchitis and emphysema)

2. Restrictive lung disease (RLD)

- ✓ This is when the chest muscles can't expand enough. This creates problems with air flow in.
- ✓ Interstitial lung fibrosis
 - ✓ Occurs when lung tissue becomes damaged and scarred

Summary of Spirometry...

Volumes	Obstructive	Restrictive
IRV	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ (high)	↓↓↓↓↓↓↓↓↓ (low)
TV	-	-
ERV	↓↓↓↓↓↓↓ (low)	↓↓↓↓↓↓↓↓↓ (low)
RV	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ (high)	↓ (Low)
TLV	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	↓↓↓↓↓↓↓↓↓

Summary of Spirometry...

Pulmonary Capacities

❖ Pulmonary Capacities – 5 types

1. The inspiratory capacity, IC

✓ $IC = TV + IRV$ & is about 3500ml

2. The expiratory capacity, EC

✓ $EC = TV + ERV$

3. The vital capacity

✓ $VC = IRV + TV + ERV$ & is about 4,500ml

4. The functional residual capacity

✓ $FRC = ERV + RV$ & is about 2300 ml

5. The total lung capacity

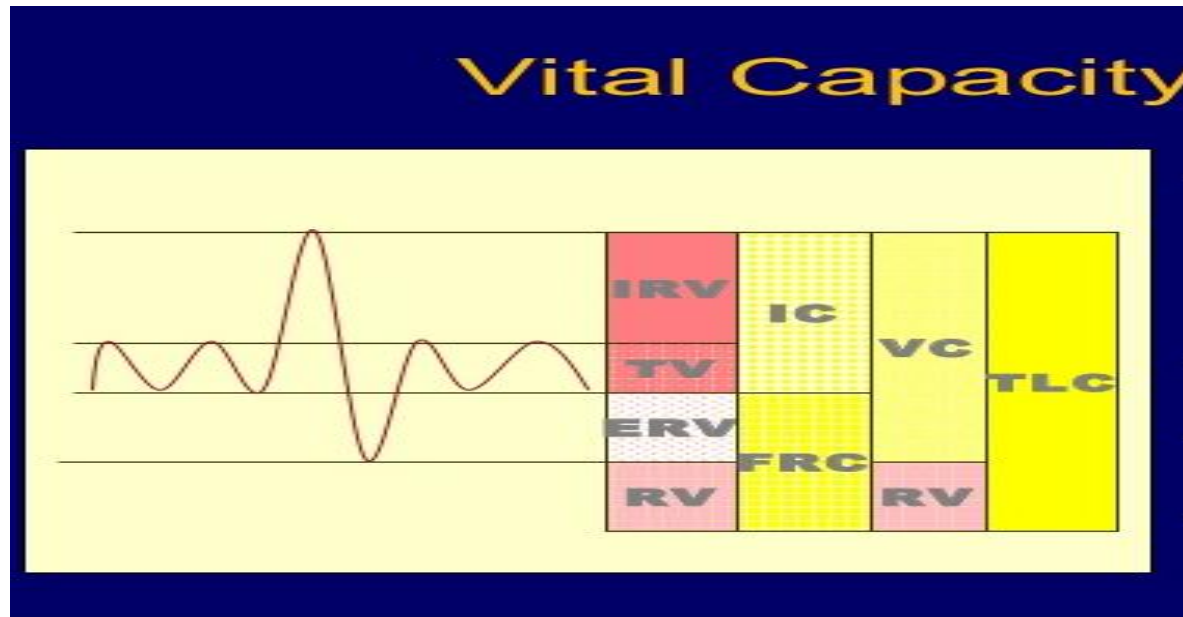
✓ $TLC = VC + RV \Rightarrow TLC = IRV + TV + ERV + RV$ & is about 5800 ml

Summary of Spirometry...

1.IC → the maximum volume of air that can be inhaled after a normal tidal expiration.

$$\text{IC} = \text{TV} + \text{IRV}$$

✓ Averages about 3500 ml



2.The expiratory capacity, EC

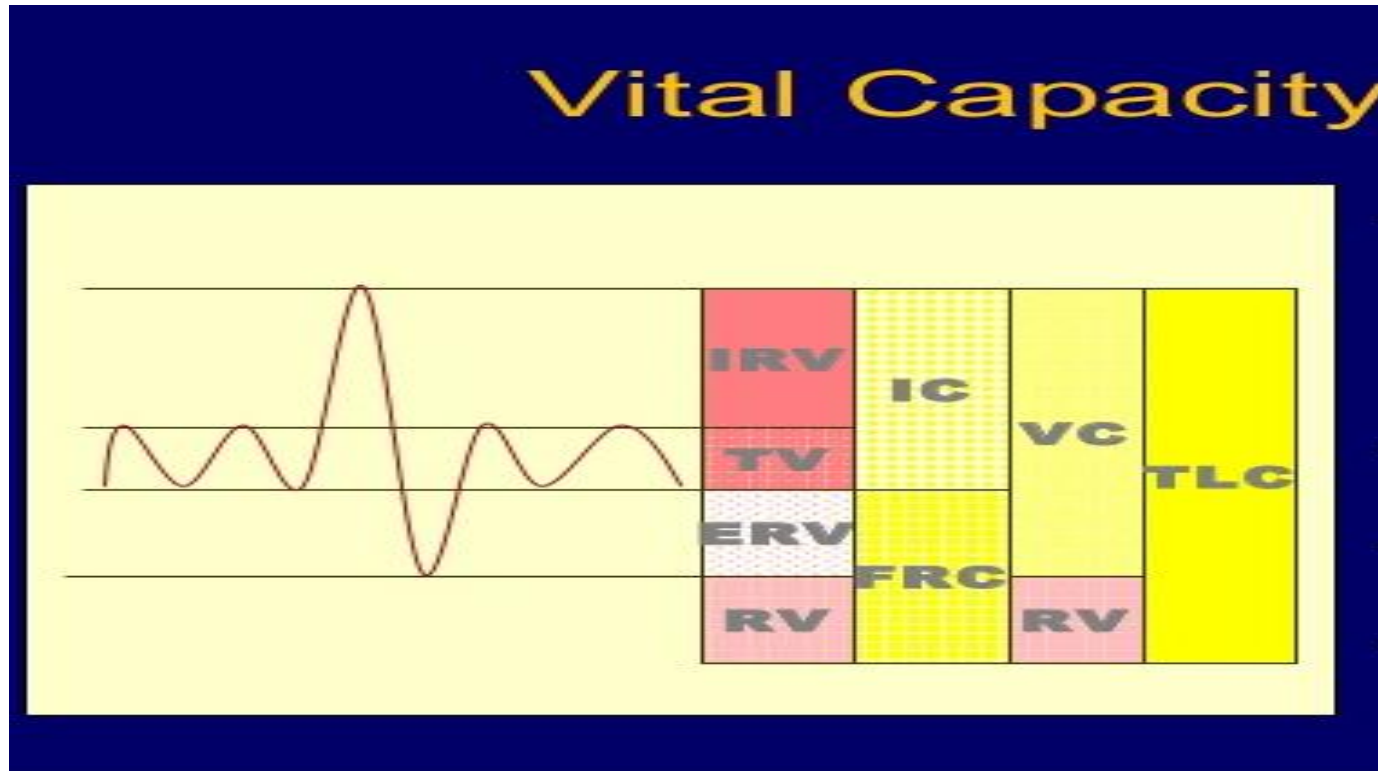
$$\text{EC} = \text{TV} + \text{ERV}$$

Summary of Spirometry...

3. Vital capacity (VC)-the max volume of air that can be expired after max inspiration.

-The total amount of exchangeable air

$$\checkmark VC = TV + IRV + ERV = 4500 \text{ mL}$$



Summary of Spirometry...

4.Functional residual capacity (FRC)-the amount of air remaining in the lung after normal tidal expiration.

✓ $FRC = RV + ERV$

✓ Averages about 2300 ml

5.Total lung capacity (TLC)-the total amount of air in the lung after max inspiration.

✓ $TLC = VC + RV = IRV + TV + ERV + RV$

✓ Averages about 5800 ml

1. Respiratory rate (RR, Bf)-is the number of breaths per minute=15 breaths/min

2. Pulmonary or minute ventilation

- ❑ The total volume of air breathed in & out from the respiratory system in 1 minute.
 - ✓ Determined by volume of air per breath & respiratory rate.
 - ✓ $PV = TV \text{ (ml/ breath)} \times \text{respiratory rate (breaths/min)} = \underline{TV \times RR}$
 - ✓ $PV = 500\text{ml/breath} \times 12 \text{ breaths/min} = \underline{6,000 \text{ ml/min.}}$
 - ✓ **6L** of air breathed in & out in one minute at rest

3. Alveolar ventilation

- ❑ Is the measure of the amount of fresh air that reaches the alveoli in each minute.
 - ✓ Is the volume of pure air participate in gas exchange b/n Atm & alveoli in a minute
 - ✓ Calculated as alveolar Ventilation = $(TV - \text{dead space vol.}) \times RR$
 - ✓ $VA = RR \times (TV - DSV)$
 - ✓ $TV = 500\text{ml}$
 - ✓ $DSV = 150\text{ml}$
 - ✓ $(500 - 150) \times 12 = \underline{4200 \text{ ml/min}}$ during quiet breathing.

Summary of Spirometry...

❑ Pulmonary ventilation (PV) is greater than alveolar ventilation (Av) Because of the presence of dead space (DS) volume in PV.

✓ DS is area of respiratory tract where there is no gas exchange

❑ 2 types of dead space

1. Anatomic dead space (150ml)

✓ 'V' of conducting passage ways of the tract where there is no gas exchange

✓ The 'V' of the nose, pharynx & tracheo-bronchial tree

2. Alveolar dead space

✓ V of any ventilated alveoli that doesn't participate in gas exchange with blood

❑ Physiological dead space = anatomical + alveolar dead spaces

✓ Under normal condition physiological dead space = anatomical dead space

✓ Because in normal condition there is no problem with the alveoli

Partial pressure

- ❑ **Atmospheric air** is a mixture of gases containing 21% O₂ and 79% N₂ with almost negligible percentages of CO₂, H₂O vapor, other gases, and pollutants.
- ❑ Altogether, these gases exert a **total atmospheric pressure of 760** mm Hg at sea level.
- ❑ **Partial pressure**: the pressure of a specific individual gas in a mixture.
 - ✓ Pressure or percentage [%] contribution of each individual gas [P_{gas}] for total [tot]

- ❑ **Dalton's law:**

$$P_{\text{gas}} = \% \text{gas} \times P_{\text{tot}}$$

- ❑ The partial pressures for oxygen (PO₂) and nitrogen (PN₂) can be calculated:
 - ✓ PO₂ = 0.21 x 760 mmHg = 160 mmHg
 - ✓ PN₂ = 0.79 x 760 mmHg = 600 mmHg

Partial pressure...

- ❑ A difference in partial pressure b/n capillary blood and surrounding structures is known as a partial pressure gradient.
- ❑ **It exists b/n :**
 - ❑ The **alveolar air** & **pulmonary capillary blood**
 - ❑ The **systemic capillary blood** & surrounding **tissues**
- ❑ A gas always diffuses **down its partial pressure gradient**.

Gas exchange or diffusion (CO_2 & O_2)

- ✓ Gas exchange at both the pulmonary capillary and the tissue capillary levels involves simple passive diffusion of O_2 & CO_2
- ✓ The diffusion of O_2 and CO_2 depends on their partial pressure gradients.
- ✓ O_2 diffuses from an area of high partial pressure in the alveoli to an area of low partial pressure in the pulmonary capillary blood.
- ✓ CO_2 diffuses down its partial pressure gradient from the pulmonary capillary blood into the alveoli.

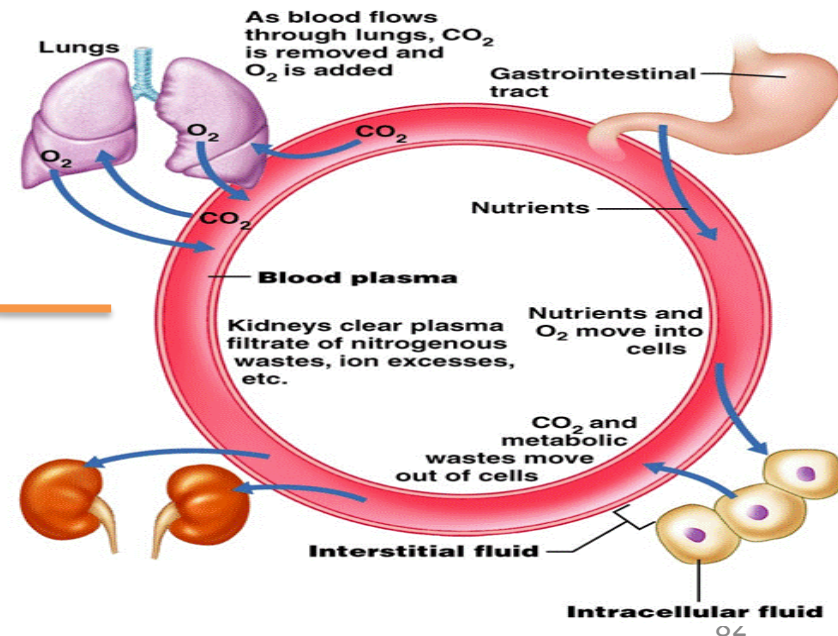
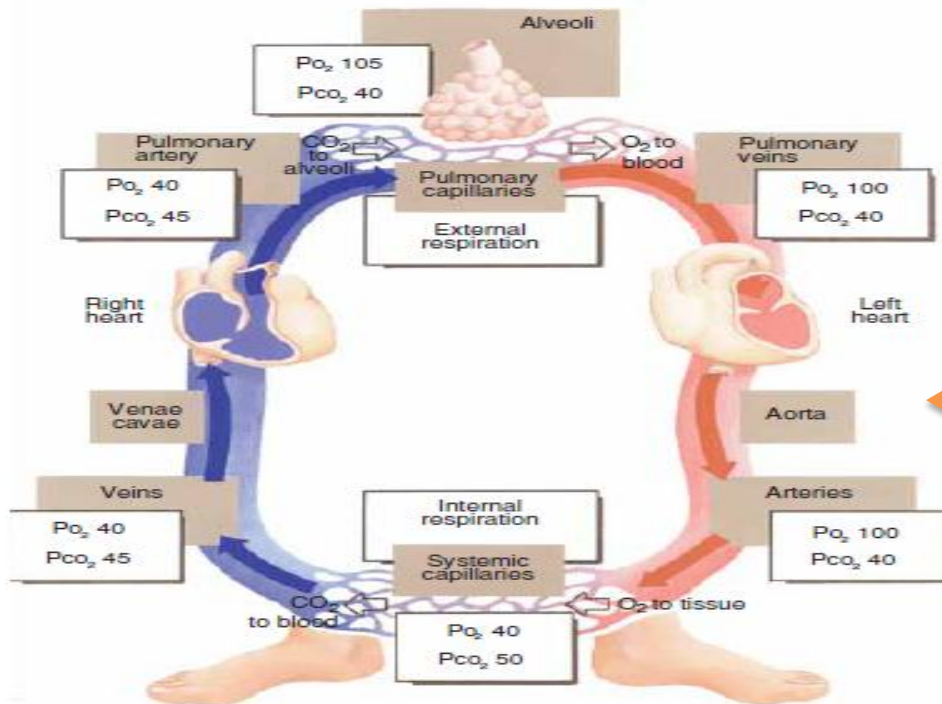
Gas exchange or diffusion...

❑ Occurs at 2 levels

1. Pulmonary-capillary → b/n pulmonary capillaries & alveoli
2. Systemic capillaries → b/n systemic capillaries & body cells

❑ Involves simple diffusion of O_2 & CO_2 down **partial pressure gradients**.

✓ No active transport mechanism involved



PO_2 & PCO_2 in the body (In normal or resting conditions)

A. In Alveoli

- ❑ $PO_2 = 105$ mm Hg
- ❑ $PCO_2 = 40$ mm Hg

B. Systemic capillaries

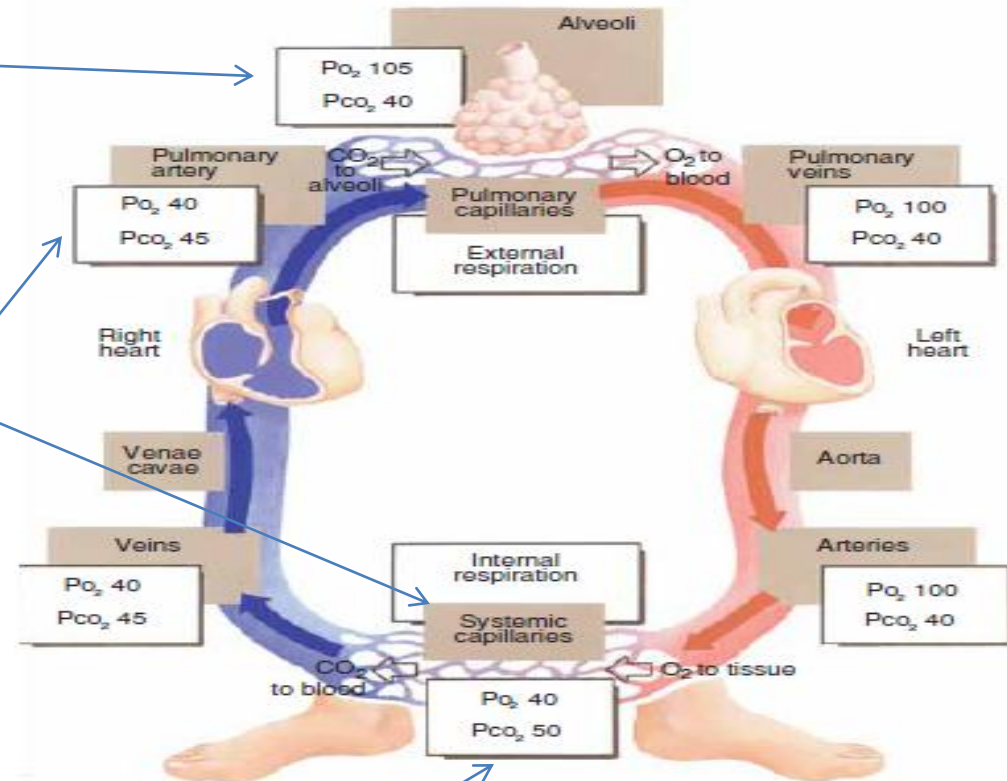
- ❑ $PO_2 = 100$ mm Hg
- ❑ $PCO_2 = 40$ mm Hg

C. In Pulmonary capillaries

- ❑ $PO_2 = 40$ mm Hg -relatively low
- ❑ Because this blood has just returned
 - ✓ From the systemic circulation & has lost much of its O_2
- ❑ $PCO_2 = 45$ mm Hg -relatively high
- ✓ Because blood returning from systemic circulation has picked up CO_2

D. Body cells (resting conditions)

- ✓ $PO_2 = 40$ mm Hg
- ✓ $PCO_2 = 45-50$ mm Hg

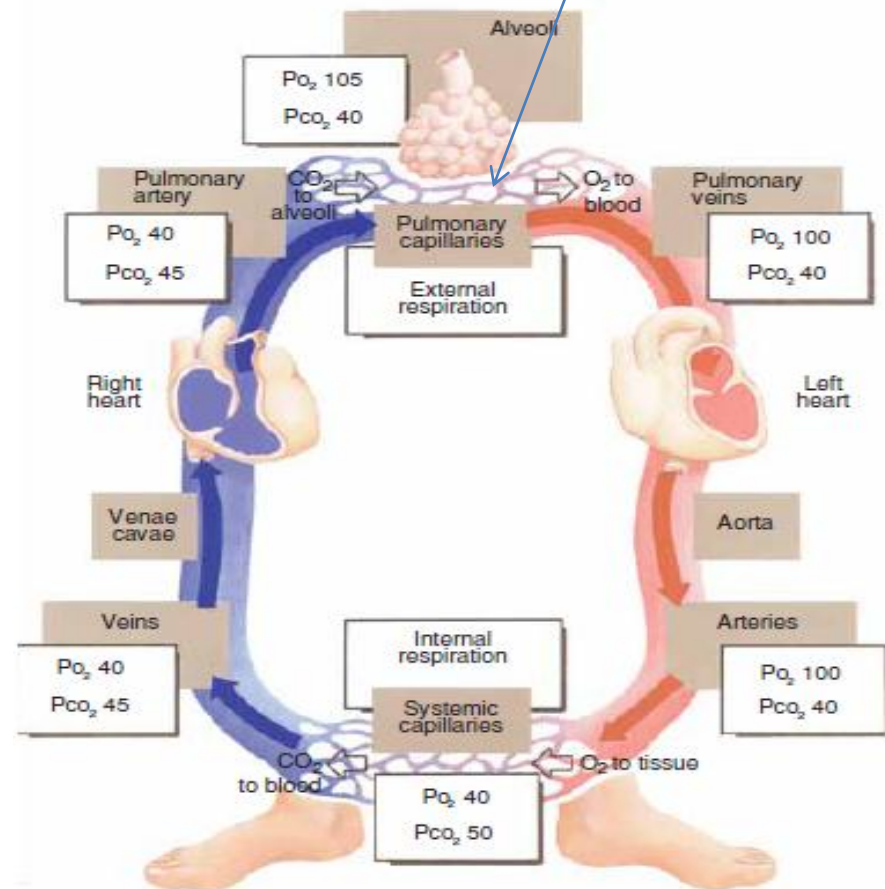
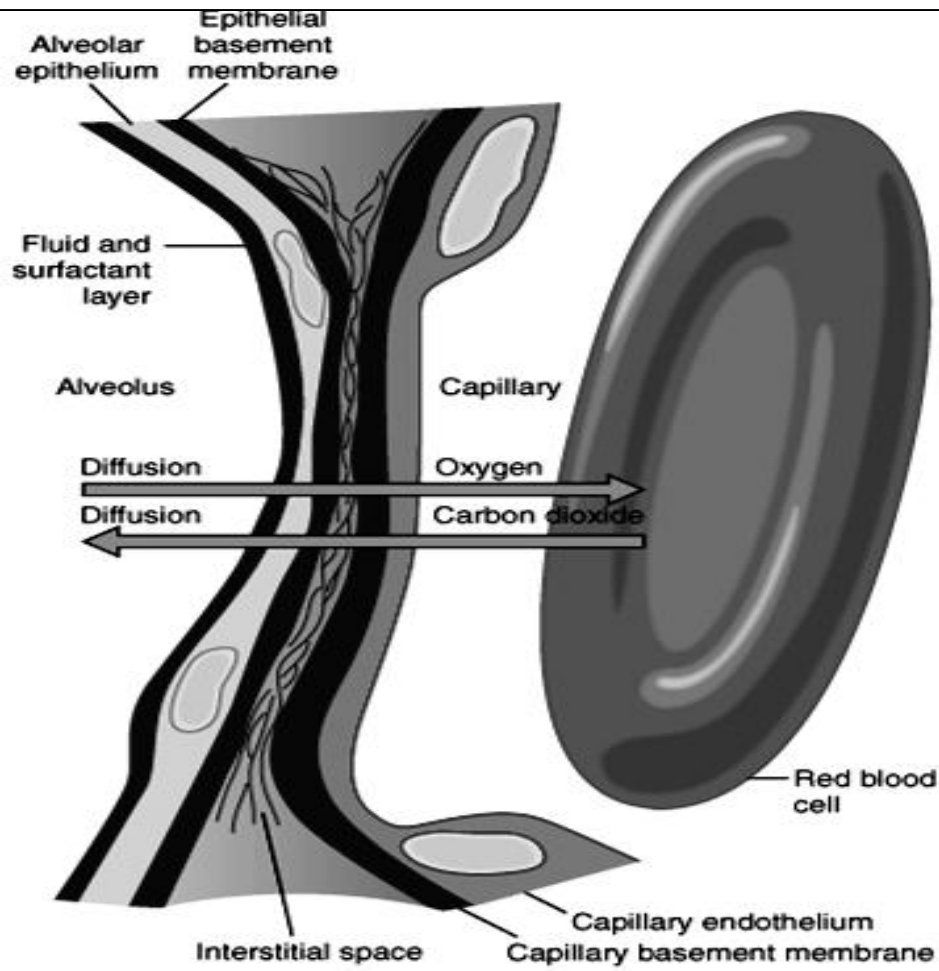


1. Gas exchange OR diffusion at the pulmonary capillaries

❑ Gases diffuse from area of high to low concentration (partial pressure)

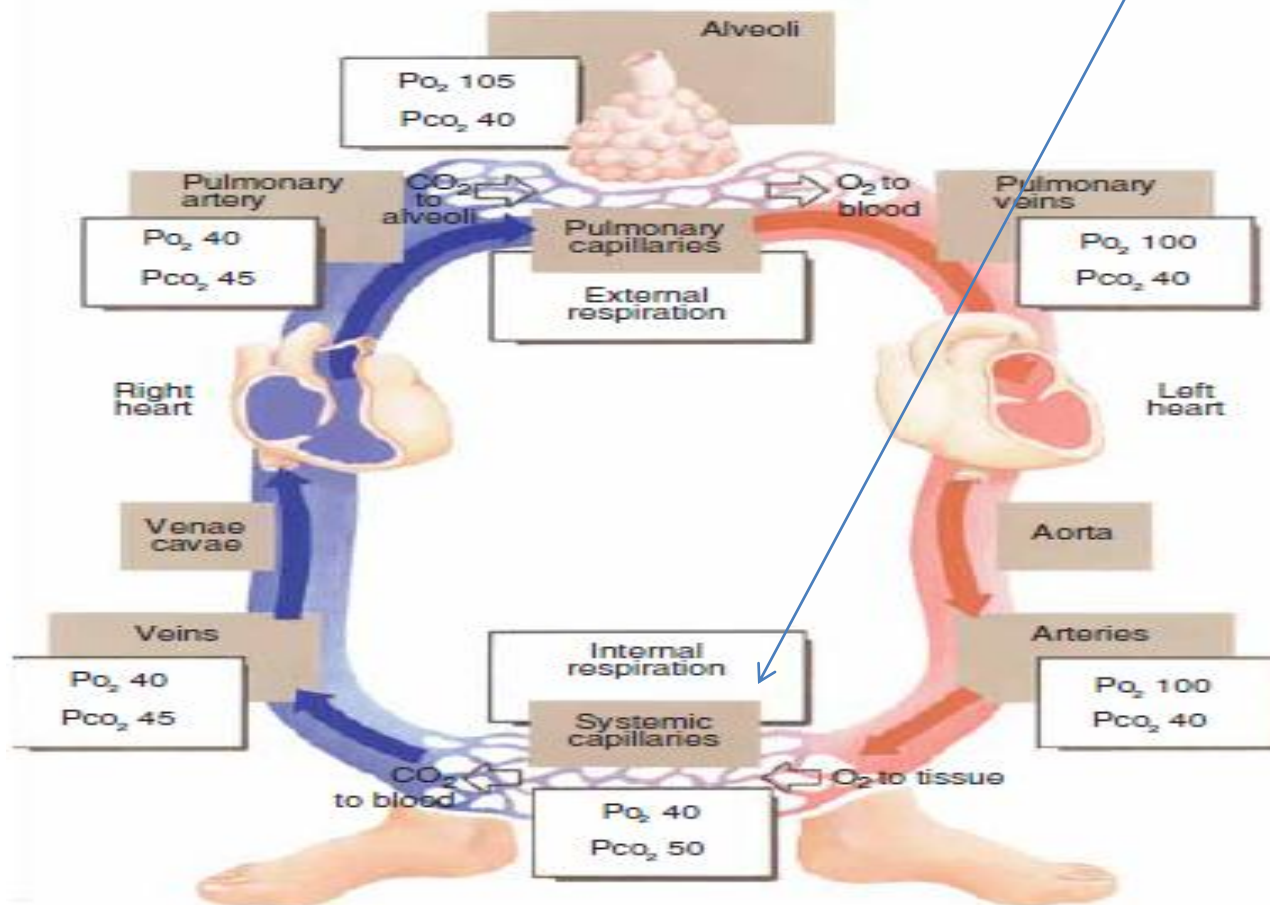
✓ O_2 from alveoli to pulmonary capillaries

✓ CO_2 diffuses from pulmonary capillaries to alveoli



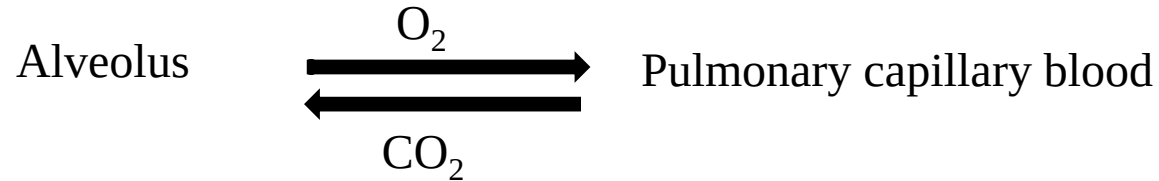
2. Gas exchange OR diffusion at systemic capillaries

- ❑ Gas exchanges b/n systemic capillaries & cells down the partial pressure gradient
 - ✓ O_2 diffuses from systemic capillaries in to body cells
 - ✓ CO_2 diffuses from cells in to the systemic capillaries

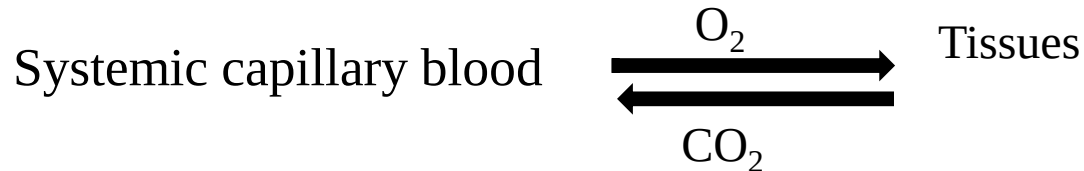


Gas exchange OR diffusion...

❖ Pulmonary gas exchange



❖ Systemic/Tissue gas exchange



Factors affecting rate of gas exchange

1. Partial pressure difference of the gases

- ✓ Rate of transfer increases as partial pressure gradient increases.

2. Surface area available for gas exchange

- ✓ Rate of transfer increases as surface area increases

3. Thickness of respiratory membrane

- ✓ Rate of transfer decreases as thickness increases

4. Diffusion (D) constant $D \propto \frac{\text{solubility}}{\sqrt{\text{MW}}}$

Molecular Weight (MW)

- ✓ O₂ has a lower molecular weight than CO₂, O₂ diffuses faster (about 1.2 times faster).

Solubility

- ✓ CO₂ is 20 x more soluble than oxygen, therefore, it diffuses more rapid than that of O₂ (20x)

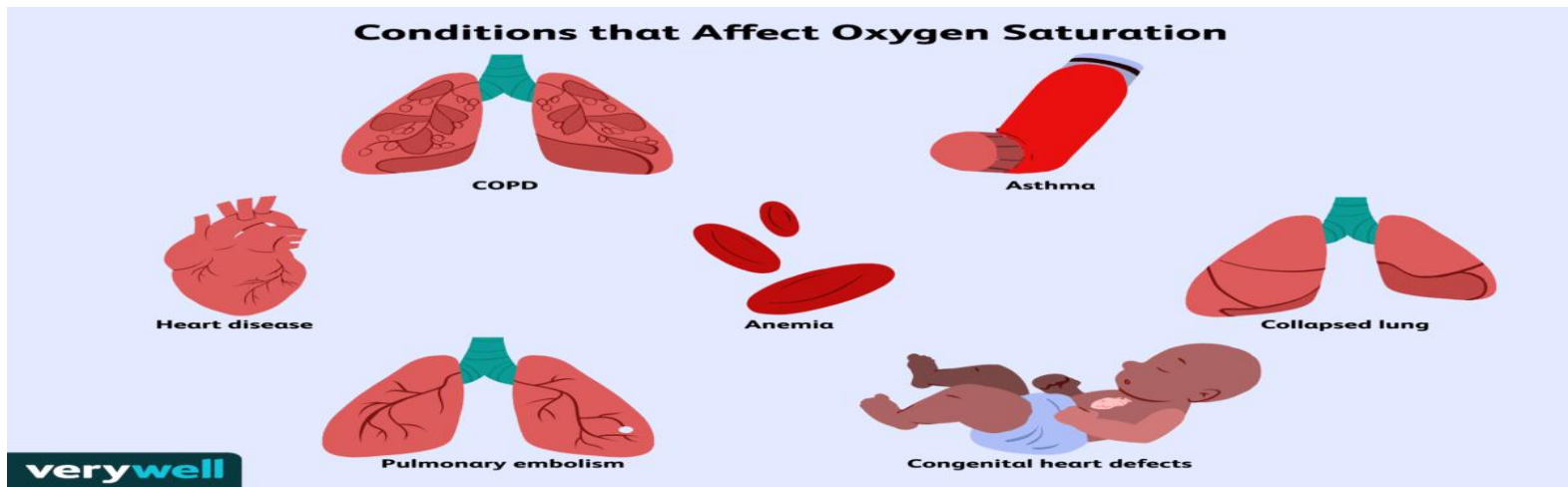
How gases [O₂ & CO₂] transported in the blood?

1. Oxygen transport via:

- a) Bound to hemoglobin (≈98%)
- b) Dissolved in the plasma (≈2%)

Hemoglobin saturation

- ☐ Is the percentage of Hgb binding sites available for binding to O₂, w/c are occupied by O₂
- ☐ Is the percentage of Hgb molecules carrying/ binding with O₂
 - ✓ For healthy 97-100% Hgb saturated or bound with O₂.



Cont'd

2. CO₂ transport in the Blood

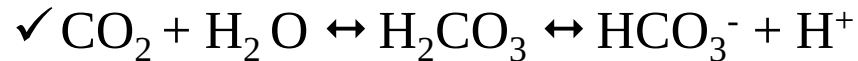
CO₂ - transported from the body cells back to the lungs as

1. Bicarbonate (HCO₃⁻) _____ 60-70%
2. Carbaminohemoglobin _____ 23-30%
3. Dissolved in the plasma _____ 7-10%

2.1 . Bicarbonate (HCO₃⁻) -60-70%

☐ Formed

- ✓ When **CO₂** (released by cells with ATP) combines with H₂O
- ✓ By action of the enzyme in RBCs called carbonic anhydrase

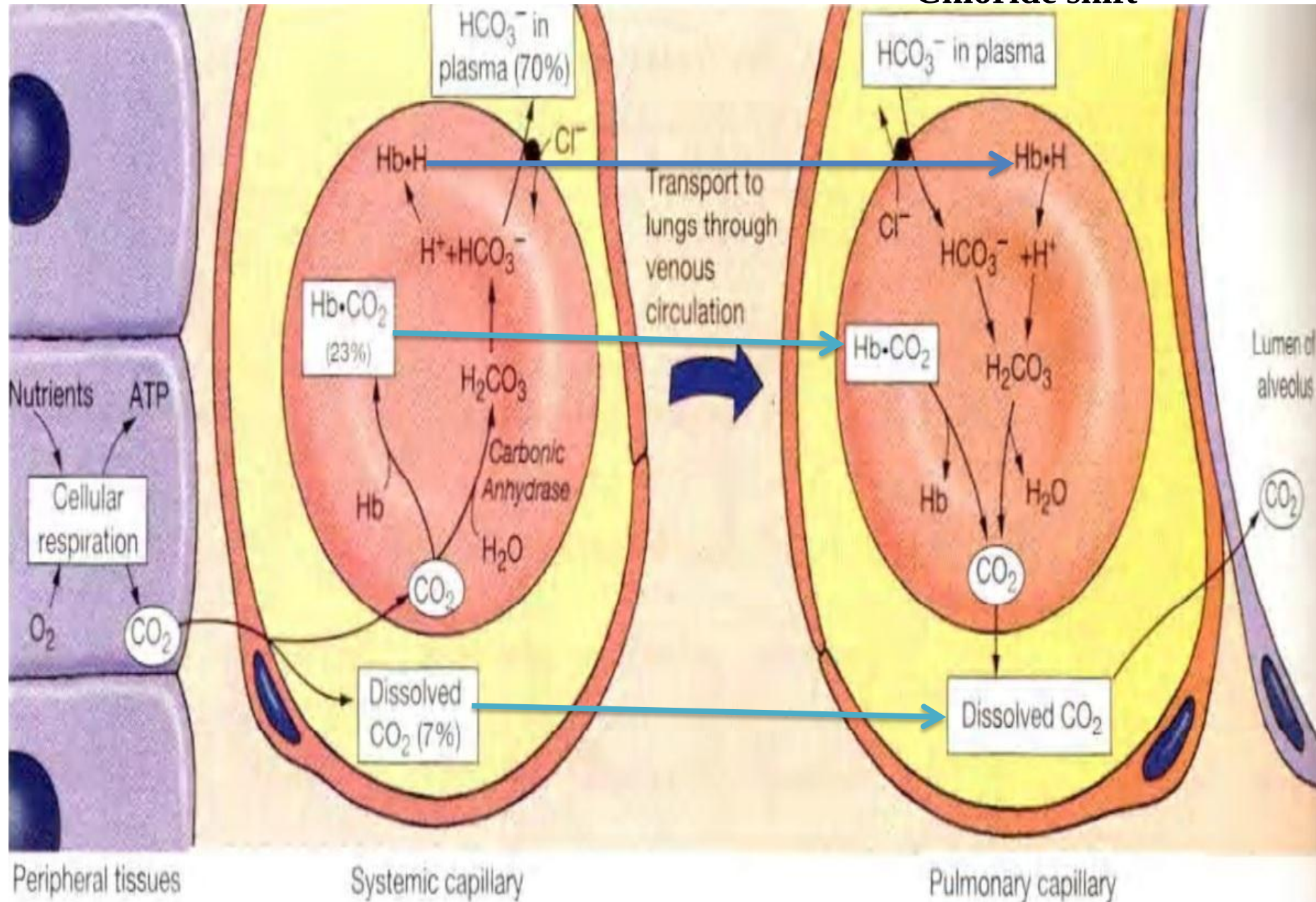


2.2 .Carbamino-hemoglobin – 23-30%

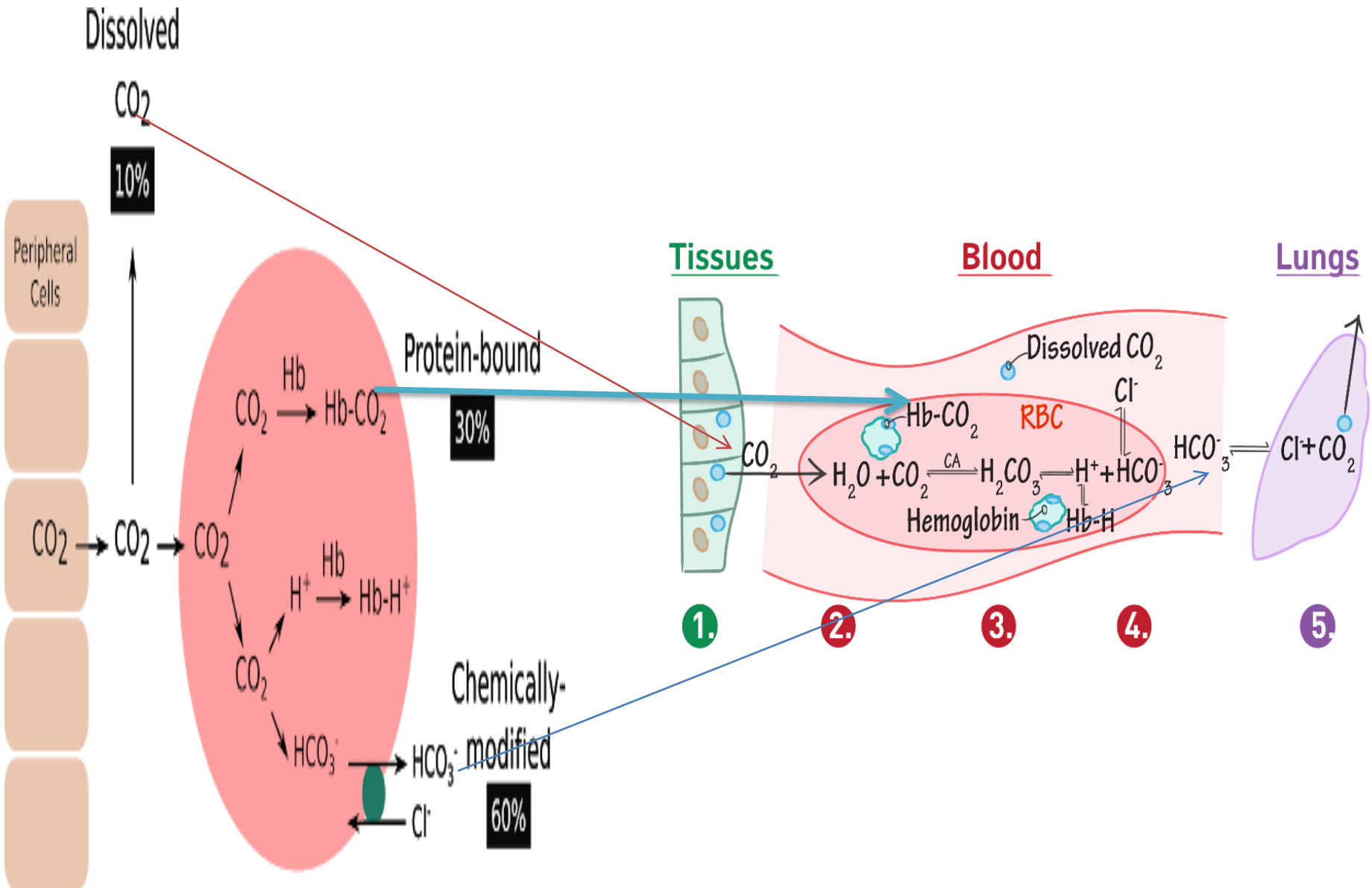
2.3 . Dissolved in the plasma -7-10%

Cont'd

Chloride shift



Cont'd



In summary

Gas Transport by The Blood

1. Oxygen is transported in the blood in 2 forms:

1- Attached in loose combination with Hgb

- ✓ Over 98% of arterial O_2 is carried in the form of oxyhemoglobin.

2- Physically dissolved:

- ✓ Less than 2% of O_2 dissolved in the arterial blood.

2. Transport of CO₂ in the blood:

❑ CO₂ is transported from body cells back to the lungs, & removed to the atmosphere

❑ CO₂ is transported in the blood in 3 forms:

1. 7-10% is dissolved in plasma

2. 23-30% is bound to the globin part of hemoglobin.

3. 60-70% is carried as bicarbonate ion in plasma.



Factors affecting transport of oxygen

1. Acidity (pH) :

- ✓ Acidity (H^+ increases, pH decreases) $\rightarrow$ decreases affinity of Hgb for $O_2 \rightarrow O_2$ dissociates more readily from Hb.

2. Carbon dioxide:

- ✓ PCO_2 increases in the systemic capillaries blood $\rightarrow$ decreases the affinity of Hgb for $O_2 \rightarrow$ Hgb unloads more O_2 at the tissue level.
- ✓ CO_2 generates H_2CO_3 , the blood becomes more acidic at the systemic capillary level.
- ✓ **Bohr effect** $\Rightarrow CO_2$ & H^+ can combine reversibly with Hgb at site other than O_2 binding site $\Rightarrow$ reduces O_2 affinity.

Factors affecting transport of ...

3. Temperature

- ✓ A rise in temperature, resulting in Hgb unloading of more O_2 .
- ✓ Exercising muscles generate heat, enhances O_2 release from Hgb for use by more active tissues.

4. Carbon monoxide (CO):

- ✓ CO is **not** a normal constituent of inspired air & It is a **poisonous** gas.
- ✓ CO and O_2 compete for the same binding sites on Hgb.
- ❑ Hgb's affinity for CO is **240 times** that of its affinity for O_2 , so it forms **carboxyhemoglobin (HgbCO)**.
- ❑ Therefore, even small amounts of CO can tie up the Hgb & **make Hgb unavailable** for O_2 transport.

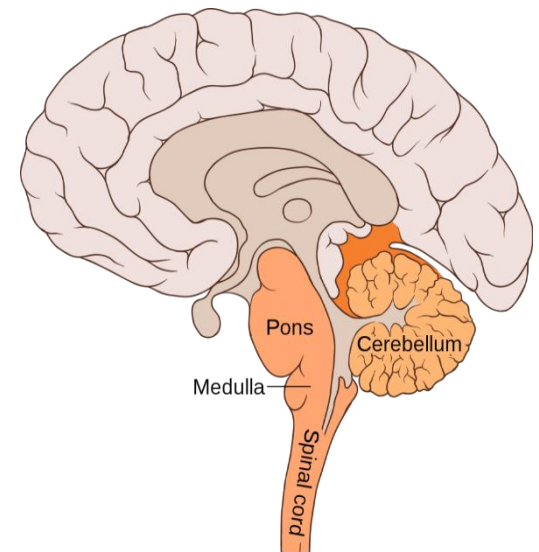
Regulation of Respiration

A. Neural regulation of respiration

1. Medullary respiratory center [DRG & VRG]
2. Pontine centers [Pneumotaxic center & Apneustic center]

B. Chemical regulations [Chemical control of ventilation]

1. Central &
2. Peripheral chemoreceptors:



Regulation of Respiration...

Respiratory control centers:

A. Medullary respiratory center

- ✓ The primary respiratory control center
- ❑ Medullary respiratory center consists of two neuronal clusters [aggregates]
 1. Dorsal respiratory group (DRG)
 2. Ventral respiratory group (VRG)
- ✓ **Provide output to the respiratory muscles**

B. Pontine centers [Pneumotaxic center and Apneustic center]

- ✓ Influence output from the medullary respiratory center

Regulation of Respiration...

❑ Dorsal respiratory group (DRG):

- ✓ In nucleus tractus solitarius; mostly inspiratory neurons, terminate on motor neurons that supply inspiratory muscles.

❑ Ventral respiratory group (VRG)

- ✓ In nucleus retroambiguus; both inspiratory & expiratory neurons
- ✓ Especially important in active expiration.
- ✓ Only during active expiration do impulses travel to expiratory muscles.

Regulation of Respiration...

Influences from pontine centers

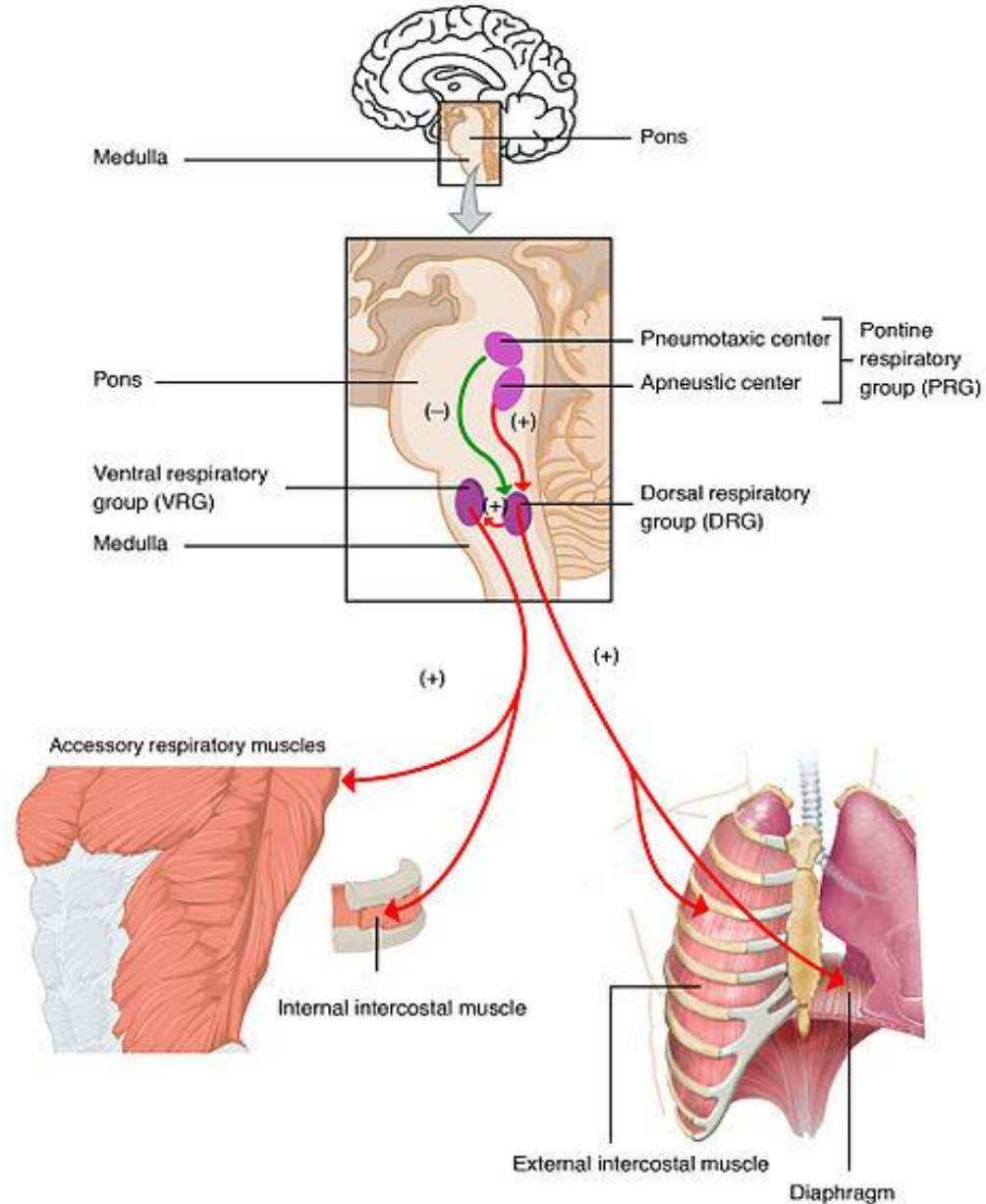
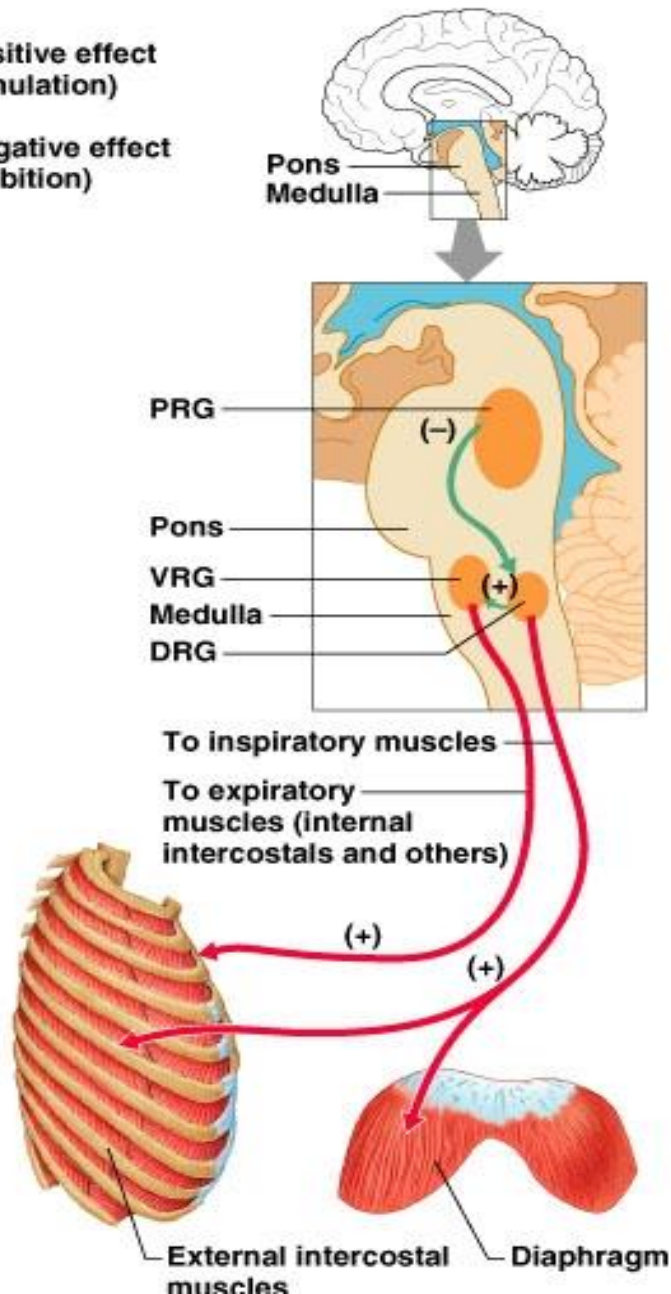
- ❑ Pontine centers exert ‘fine tuning’ influences on medullary center → ensure normal, smooth breathing, influence timing of switching between inspiration & expiration.
- 1. **Pneumotaxic center** (upper pons) sends impulses to DRG to ‘turn off’ inspiratory neurons, limiting the duration of inspiration.
- 2. **Apneustic center** (lower pons) prevents inspiratory neurons from being turned off, providing an extra boost to the inspiratory drive.
- ❖ Of the two, the pneumotaxic center dominates.
- ❖ Without pneumotaxic ‘brakes’, **apneusis**= prolonged inspiratory breaths with very short interrupting of expirations.

Regulation of Respiration...

Key:

(+) = Positive effect
(stimulation)

(-) = Negative effect
(inhibition)



Regulation of Respiration...

B. Chemoreceptors

- ❑ Provide the **most important** input to the medullary respiratory center
- ❑ Are sensitive to changes in **PO₂**, **PCO₂**, and **pH**.
- ❑ **Two types:** Peripheral chemoreceptors & Central chemoreceptors

1. Peripheral chemoreceptors

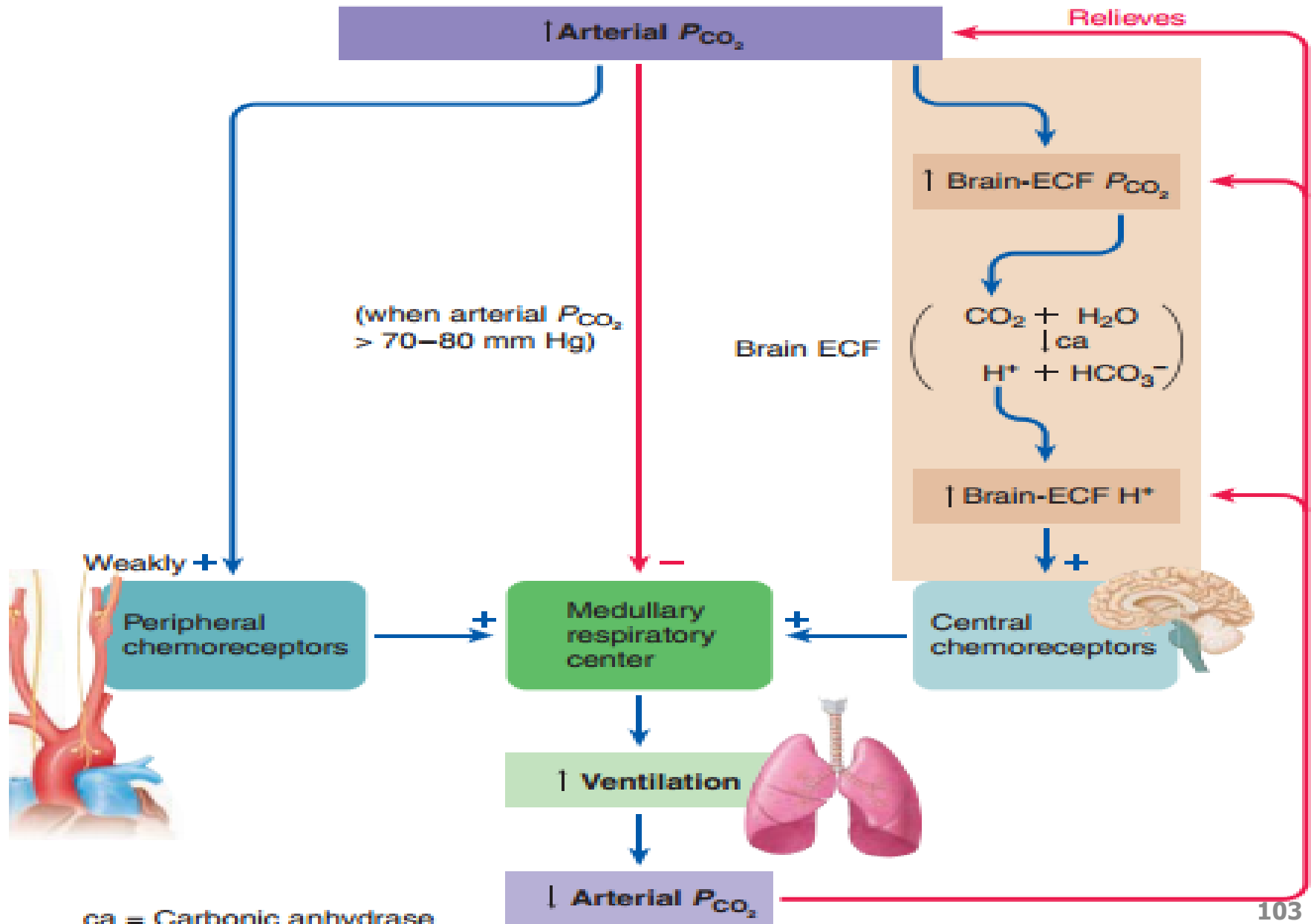
- ✓ Carotid bodies, aortic bodies respond **primarily to decreases in PO₂**
- ✓ When PaO₂ falls below 100 mmHg, **rapid** response; below 60 mmHg– **dramatic** response (breathing rate increases)
- ❖ **Site:** in the carotid & aortic bodies
- ❖ **Stimuli:** ↓ in arterial PO₂, ↑PCO₂, ↓PH
- ❖ **Stimulation:** send stimulatory impulses to Medulla [rhythmicity center =DRG].

Regulation of Respiration...

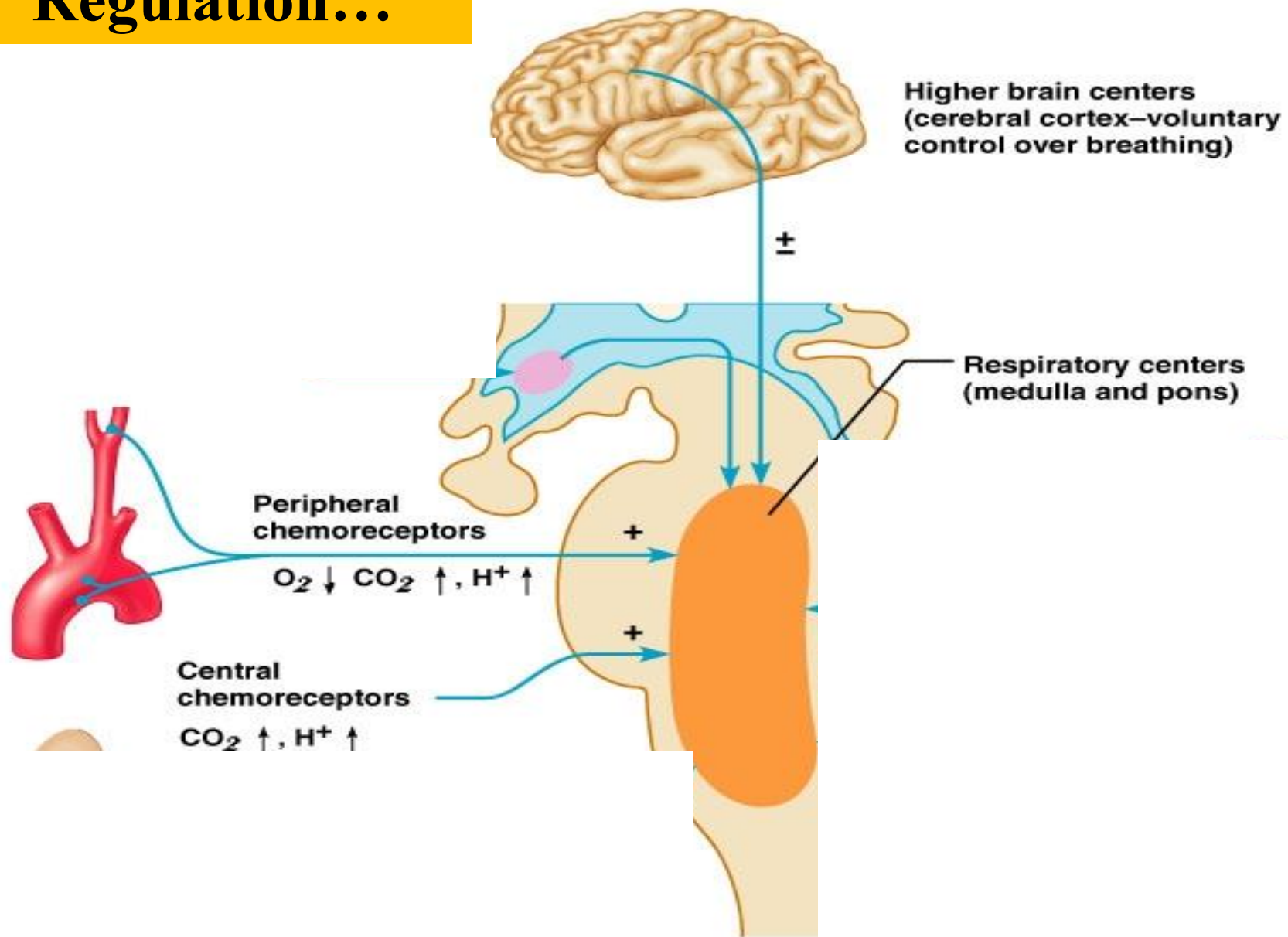
2. Central chemoreceptors

- ✓ Near the ventral surface of the medulla in close proximity to the respiratory center.
- ✓ Are sensitive to the pH of the cerebrospinal fluid (CSF)
- ✓ H^+ acts directly on the central chemoreceptors
- ✓ Thus, increases in PCO_2 and $[H^+]$ stimulate breathing (hyperventilation), and decreases in PCO_2 and $[H^+]$ inhibit breathing (hypoventilation).
- ✓ **Site:** medulla
- ✓ **Stimuli:** H^+ ion concentration in the CSF
- ✓ H^+ ion can not cross the blood brain barrier, but it increases in the CSF secondary to $\uparrow PCO_2$ in the blood, which pass through BBB to the CSF
- ❑ It sends stimulatory impulse to stimulates ventilation

Regulation of Respiration...



Regulation...



Clinical Correlates

A. Abnormalities in arterial PO_2

❑ **Hypoxia:** the condition of having **insufficient** O_2 in blood [tissues are starved of O_2].

❑ **There are 4 general categories of hypoxia:**

1. Hypoxic hypoxia (Hypoxemia)

✓ Is a decrease in arterial PO_2 ➔ inadequate Hgb saturation

✓ **Caused by:**

1) Inadequate gas exchange

2) Exposure to high altitude or to a suffocating environment

Clinical Correlates

2. Anemic hypoxia

- ❑ A reduced O_2 - carrying capacity of the blood.
- ❑ It can result from:
 - ✓ A decrease in circulating RBCs
 - ✓ Inadequate amount of Hgb
 - ✓ CO poisoning
- ❑ The arterial PO_2 is normal but the O_2 content of the arterial blood is lower than normal because of the reduction in available Hgb or RBC.

Clinical Correlates

3. Circulatory Hypoxia

- ❑ Is decreased O₂ delivery to the tissues.
- ✓ Can be restricted to a limited area by a local vascular spasm or blockage.
- ✓ The body may experience circulatory hypoxia in general, from congestive heart failure or circulatory shock.
- ✓ Arterial PO₂ and O₂ content are typically normal

Clinical Correlates

4. Histotoxic hypoxia

- ✓ O₂ delivery to the tissues is normal, but the cells cannot use the O₂ available to them.
 - ✓ The classic example is cyanide poisoning.
 - ✓ Cyanide blocks cellular enzymes essential for cellular respiration (enzymes in the electron transport system).
- ❑ **Hyperoxia** : an above-normal arterial PO₂ ➡ O₂ toxicity ➡ damage some cells.
- ✓ In particular, brain damage and blindness-causing damage to the retina are associated with O₂ toxicity.

Clinical Correlates

B. Abnormalities in arterial PCO_2

❑ **Hypercapnia:**

- ✓ The condition of having excess CO_2 in arterial blood
- ✓ Caused by hypoventilation

❑ **Hypocapnia:-** below-normal arterial PCO_2 levels, is brought about by hyperventilation.

❑ **Hyperventilation occurs when a person “over breathes”.**

- ✓ Rate of ventilation $>$ body's metabolic needs ➡ CO_2 is blown off to the atm.
 - ➡ arterial PCO_2 falls
- ✓ Can be triggered by anxiety states, fever, and aspirin poisoning.

Clinical Correlates

C. Consequences of abnormalities in arterial blood gases

1. **Respiratory acidosis** is the result of abnormal CO_2 retention arising from hypoventilation (increase in CO_2 generates more H^+).
 - ✓ **Causes** : lung disease, depression of the respiratory center by drugs or disease, nerve or muscle disorders that reduce respiratory muscle ability.
2. **Respiratory alkalosis** is the result of excessive loss of CO_2 from the body as a result of hyperventilation.
 - ✓ Pulmonary ventilation increases out of proportion to the rate of CO_2 production ➡ too much CO_2 is blown off ➔ less $[\text{H}^+]$ is formed.

Clinical Correlates

D. Others

- ❑ **Pnea** ➡ breath
- ❑ **Eupnea** ➡ Normal breathing [Normal, good, relaxed, unlabored breathing.
- ❑ **Apnea** ➡ Transient cessation of breathing
 - ✓ **Respiratory arrest**: Permanent cessation of breathing (unless clinically corrected)
- ❑ **Dyspnea** ➡ Difficult or labored breathing (feeling of air hunger)
- ❑ **Asphyxia** ➡ O₂ starvation of tissues, caused by a lack of O₂ in the air, respiratory impairment, or inability of the tissues to use O₂.
- ❑ **Cyanosis** ➡ a bluish or purplish discoloration (as of skin) due to deficient oxygenation of the blood]
- ❑ **Hyperpnea** ➡ abnormally deep or rapid respiration.
- ❑ **Hypopnea** ➡ Abnormally slow, shallow breathing
- ❑ **Apneusis** ➡ prolonged inspiratory

Clinical Correlates

- ❑ Air, blood, pus & H₂O in the pleural cavity will increase interpleural pressure
 1. **Pneumothorax** ➡ when air accumulates in the pleural cavity
 2. **Hydrothorax** ➡ when water accumulates in the pleural cavity
 3. **Heamothorax** ➡ accumulation of blood in the pleural cavity
 4. **Pyothorax** ➡ accumulation of pus in the pleural cavity

The End.